

UAV Modelling and Control

Instructor:

Prof. Chen Xi (Patrick), ERB310, Tel: 39430814, xichen002@cuhk.edu.hk

Tutor:

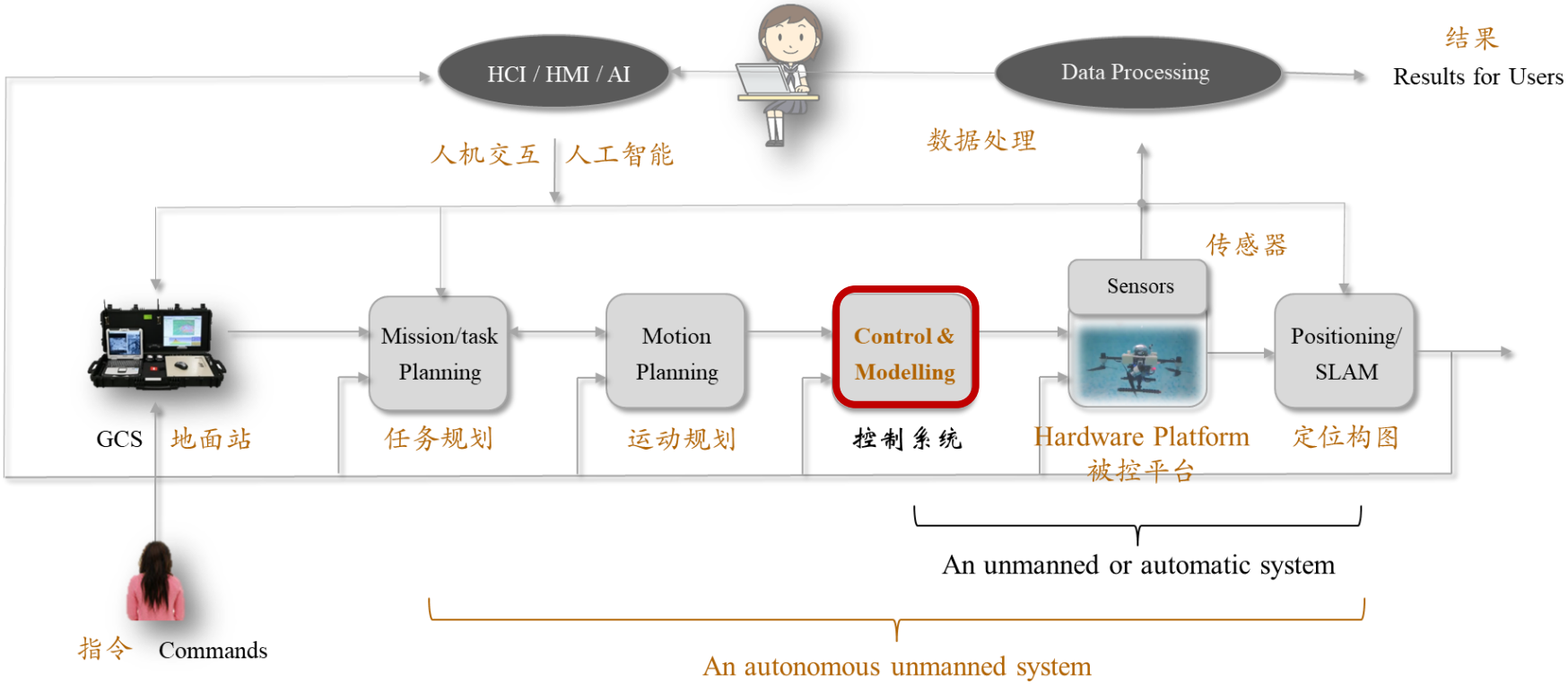
Zhipeng Guo, zhguo2@mae.cuhk.edu.hk

Haochen Hong, hchong@link.cuhk.edu.hk

Yiding Liu, ydliu@mae.cuhk.edu.hk

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Dynamic modelling



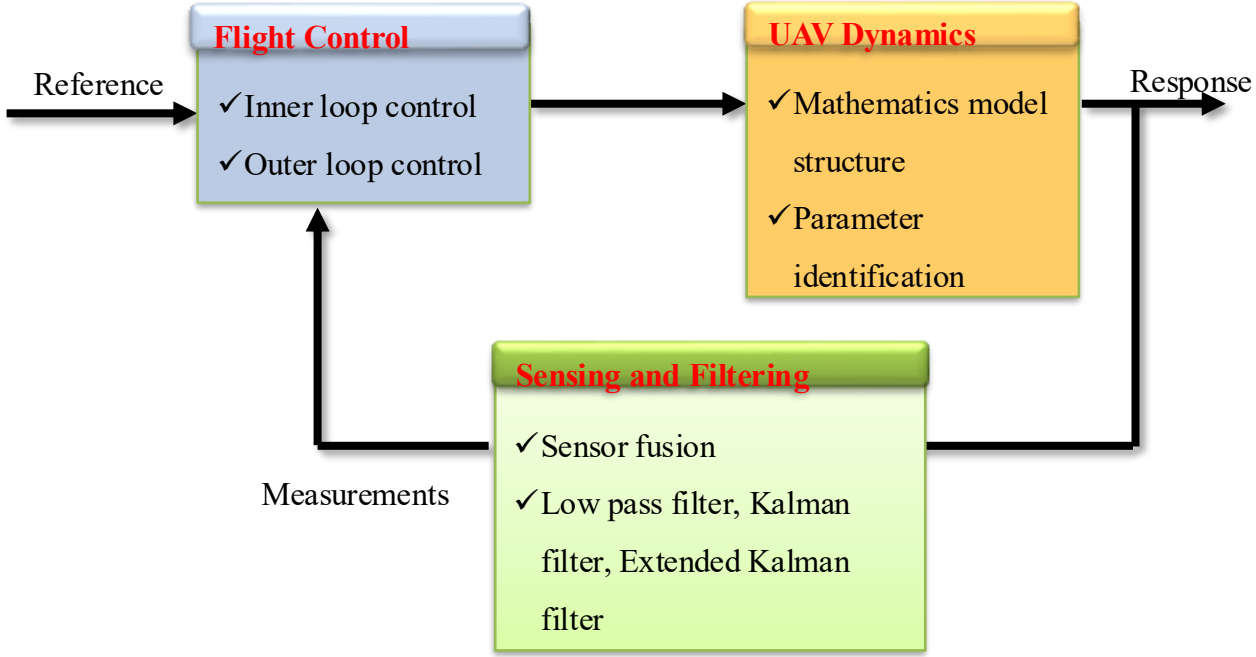
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Outline

- Introduction - UAV system overview
- Flying theory of multi-rotor UAVs
- Mathematics modeling
- Parameters identification
- Model verification

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Introduction – System Overview



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Flying Theory of Multi-rotor UAVs

Different Types of UAV – Why multi-rotor is preferred?

Fixed-wing



Source: 3dr.com

Single-rotor (helicopter)



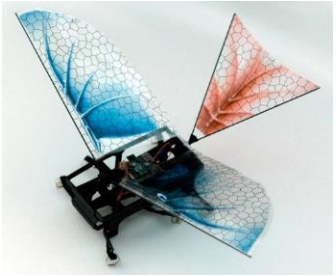
Source: arstechnica.com

Co-axial helicopter



Source: diydrones.com

Flapping-wing



Source: blogs.bu.edu

Multi-rotor



Source: service-drone.com

Others (unconventional)



Why?

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Flying Theory of Multi-rotor UAVs

Capabilities Comparison

	VTOL	Mechanical simplicity	Control simplicity	Scale	Payload
Fixed-wing	No	Simple	Simple	Medium-Large	High
Single-rotor	Yes	Complex	Complex	Medium-Large	High
Co-axial	Yes	Complex	Complex	Small-Medium	Medium
Multi-rotor	Yes	Simple	Simple	Small-Medium	Medium
Flapping-wing	Yes	Complex	Complex	Small	Low

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Flying Theory of Multi-rotor UAVs

Different Types of Multi-rotor

Quad-rotor (quadcopter)



Hexa-rotor (hexacopter)



Octa-rotor (octacopter)



Y6



X8



1. What is the minimum or maximum number of rotors?

2. How should the rotors be arranged?

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Flying Theory of Multi-rotor UAVs

How a Multi-rotor moves?

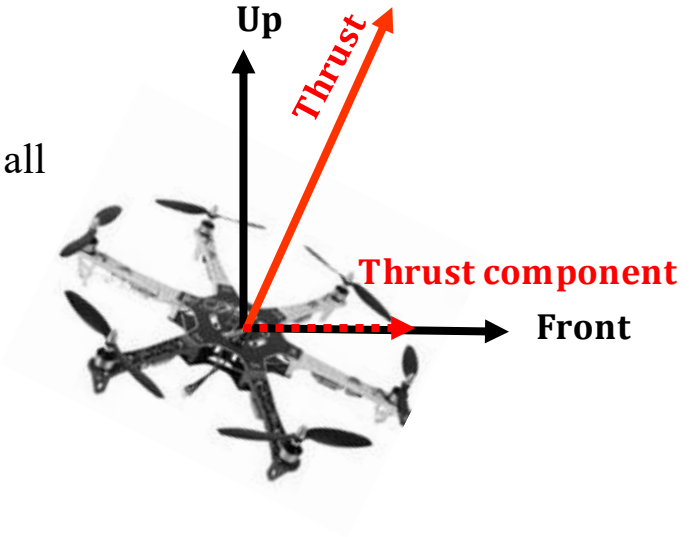
Four control inputs (channels)

Throttle : Main thrust of the UAV, produced evenly by all rotors

Aileron : Rolling movement of the UAV

Elevator : Pitching movement of the UAV

Rudder : Yawing (heading) movement of the UAV



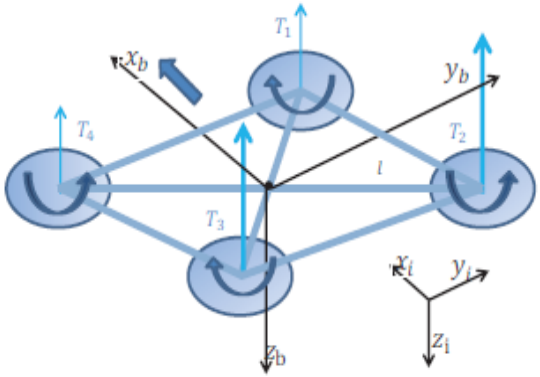
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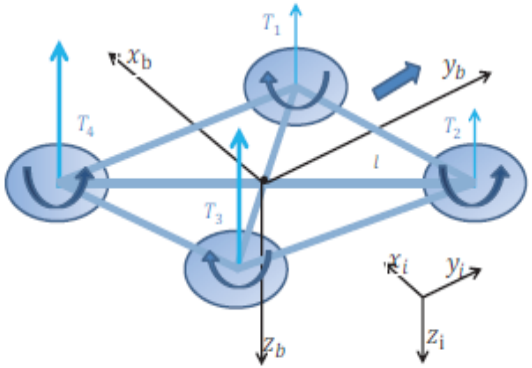
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Flying Theory of Multi-rotor UAVs

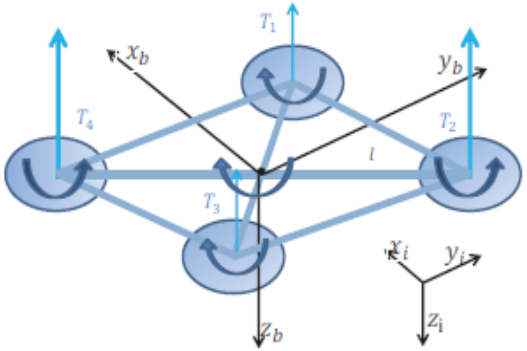
How a Multi-rotor moves?



(a) Pitching



(b) Rolling



(c) Yawing

Underactuated?

Q1. What is the minimum/maximum number of rotor?

If you need it to have 4 DoF, you'll need at least 4 rotors (static, fixed-pitch)

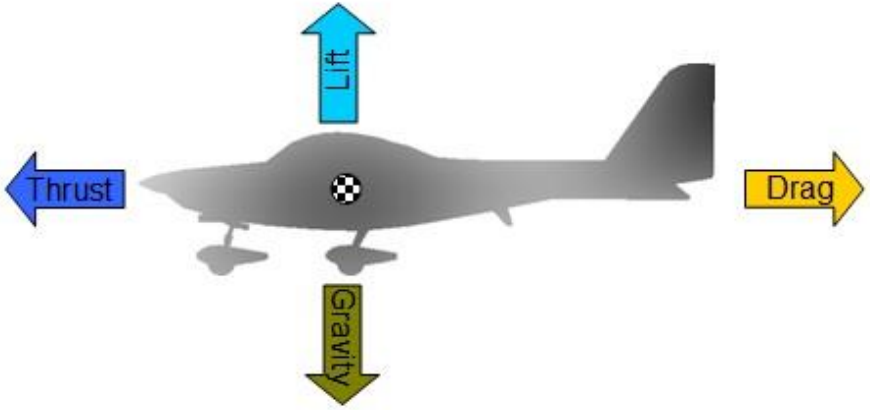
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Mathematics Modeling

What is a **flight dynamics model**?

A flight dynamics model is a set of equations that explains

1. How aerodynamic forces are generated on the aircraft body?
2. How **actuators** and external factors affect these forces?
3. How these forces affect the aircraft's motion?



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Mathematics Modeling

Why do we need a **flight dynamics model?**

1. **Design and tuning of autonomous flight control law**

Modern control techniques are mostly model based. An accurate flight dynamics model can assist flight control law design efficiently in terms of parameter tuning and performance evaluation.

2. **Design and modification of aircraft structure**

Aerodynamic stability, physical limits and characteristics can be analyzed based on the obtained model. Mechanical modification or re-structuring can be applied if there is inherent deficiencies in the original structure.

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Mathematics Modeling

How do we obtain a **flight dynamics model?**

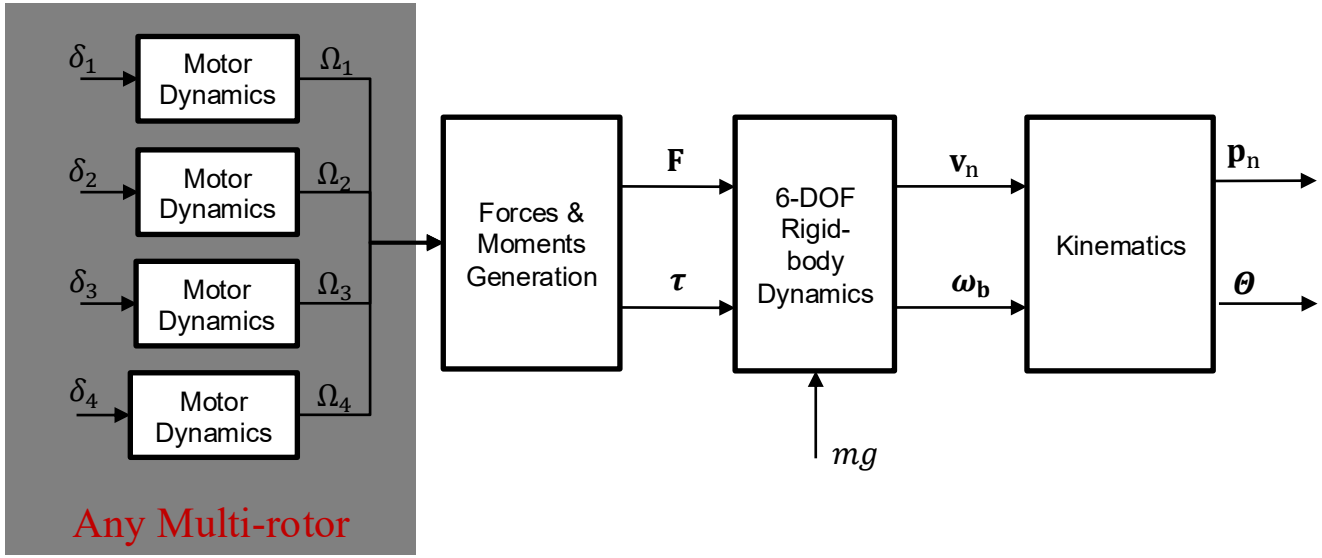
1. **System Identification Approach**
   - Build mathematical model by fitting measurement data
   - Simplified, linear, black-box
   - Flight test experiments
2. **First-principles Modeling Approach**
   - Build mathematical model by manipulating equations of physics
   - Complex, nonlinear, rigorous with physical meanings
   - Test-bench experiments
3. **Combination of the above two approaches**

Review: which part of the drone is test-bench usually applied to?

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Mathematics Modeling

First principle approach build mathematical model with strong physical meaning. Nonlinearity can also be reflected in the model. A quadrotor UAV is used here as an illustrative example.



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Mathematics Modeling

Assumptions

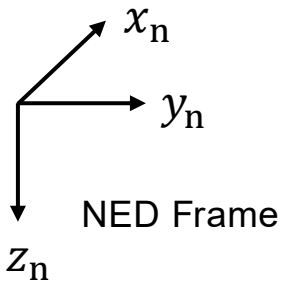
- Assumption 1. The multirotor UAV is a rigid body.
- Assumption 2. The mass and the moments of inertia are constant.
- Assumption 3. The geometric center and the Center of Gravity (CoG) of the multirotor UAV are the same.
- Assumption 4. The multirotor UAV is only under gravity and propeller thrust. Furthermore, the gravity points along the positive direction of the axis and the propeller thrust points along the negative direction of the axis.
- Assumption 5. Propellers with odd indices rotate anticlockwise and propellers with even indices rotate clockwise.

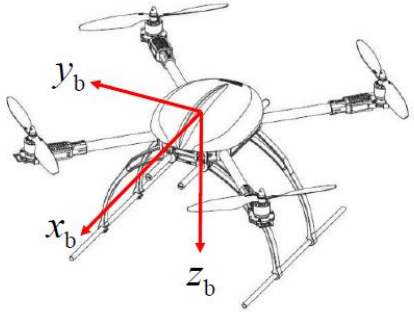
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Mathematics Modeling

Frames Assignment (Right Hand Rule):

1. **NED frame** – In UAV control, commands are usually given based on global frame (world's perspective) as operators are on the ground.
2. **Body frame** – Some sensors attached on the UAV's body gives measurement on UAV's perspective; motor thrusts are fixed on body's frame.





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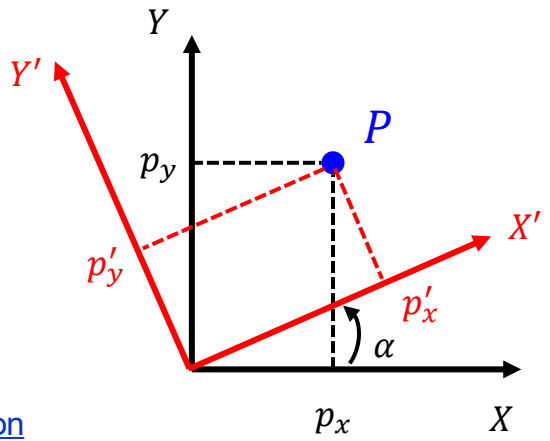
Mathematics Modeling

Frames Transformation:

- Suppose the coordinates, (p_x, p_y), of a point P in the two-dimensional XY system are known, but we want to know the coordinates of this point in another coordinate system, $X'Y'$, which is related to the XY system by a counter-clockwise rotation by an angle α.
- We can calculate the coordinate of P with respect to frame $X'Y'$ as

$$\begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = \begin{bmatrix} p_x \cos\alpha + p_y \sin\alpha \\ -p_x \sin\alpha + p_y \cos\alpha \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$\begin{bmatrix} p'_x \\ p'_y \end{bmatrix} = R(\alpha) \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$



Properties of Rotation Matrix

- $R(-\alpha) = \big(R(\alpha)\big)^T = \big(R(\alpha)\big)^{-1}$

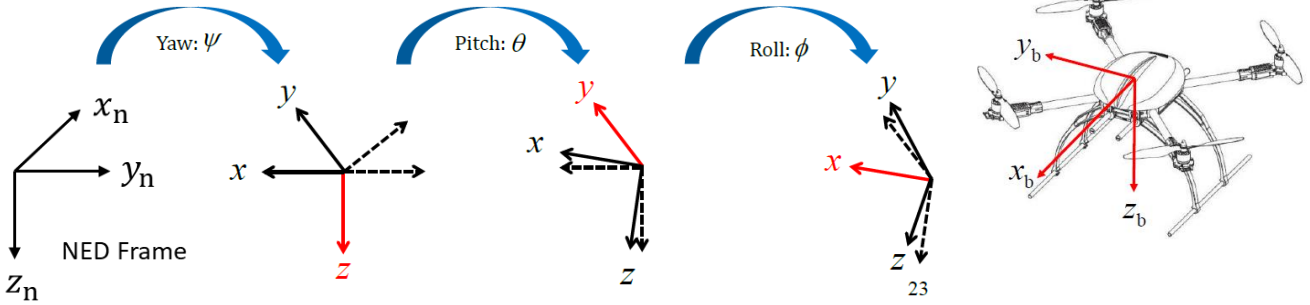
Orthogonal matrix

http://anorganik.uni-tuebingen.de/klaus/nmr/index.php?p=conventions/euler/rotation

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Mathematics Modeling

Frames Assignment:



$$\mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

$$\mathbf{R}_{b/n} = \mathbf{R}_x(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\psi) = \begin{bmatrix} c_\theta\, c_\psi & c_\theta\, s_\psi & -s_\theta \\ s_\phi s_\theta c_\psi - c_\phi s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi c_\theta \\ c_\phi s_\theta c_\psi + s_\phi s_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi & c_\phi c_\theta \end{bmatrix}$$

Roll Pitch Yaw

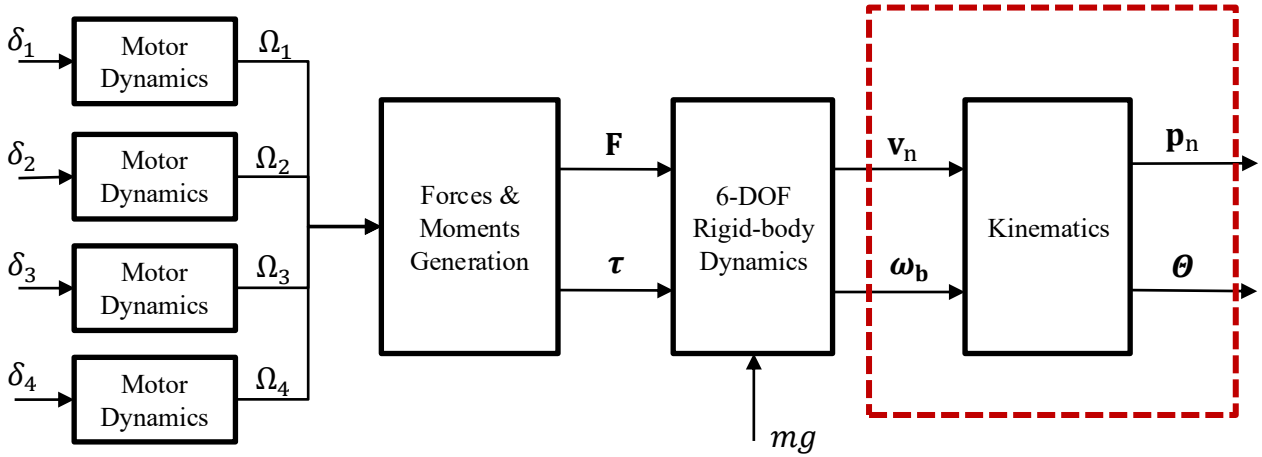
$$\mathbf{R}_{n/b} = \mathbf{R}_{b/n}^T = \begin{bmatrix} c_\theta\, c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta\, s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}$$

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Mathematics Modeling

Kinematics:

Classical mechanics which describes the **motion** of points, bodies (objects) and systems of bodies (groups of objects) without consideration of the causes of motion.



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Mathematics Modeling

Kinematics:

Classical mechanics which describes the **motion** of points, bodies (objects) and systems of bodies (groups of objects) without consideration of the causes of motion.

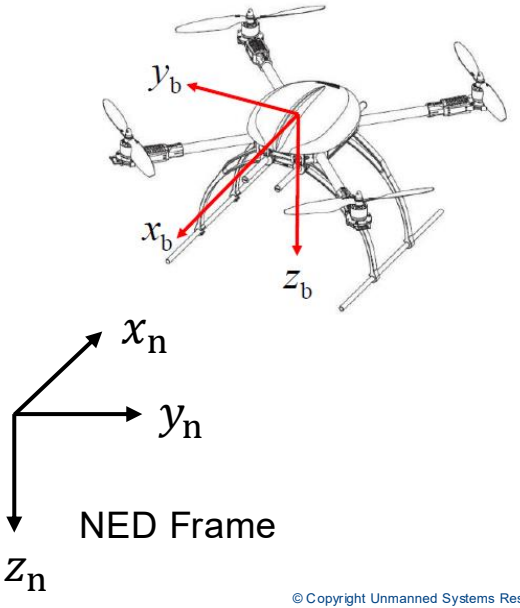
$$\dot{\mathbf{p}}_{\mathrm{n}} = \mathbf{v}_{\mathrm{n}}$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}$$

where

$$\boldsymbol{\Theta} \triangleq \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} \qquad \mathbf{W} = \begin{bmatrix} 1 & \tan\theta \sin\phi & \tan\theta \cos\phi \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi / \cos\theta & \cos\phi / \cos\theta \end{bmatrix}$$

W: Body frame to Euler Angles



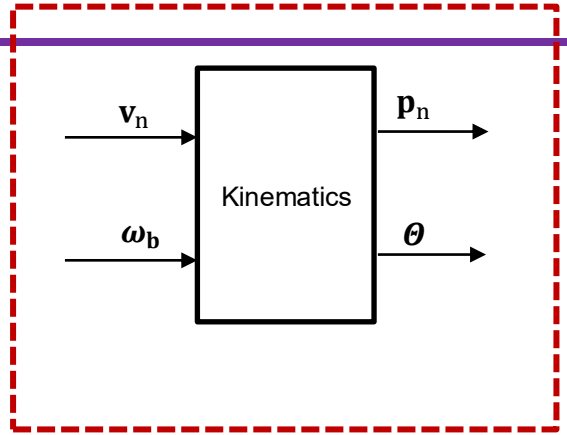
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Mathematics Modeling

Kinematics:

$$\dot{\mathbf{p}}_{\mathrm{n}} = \mathbf{v}_{\mathrm{n}}$$

$$\dot{\Theta} = \mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}$$



Derivative of NED frame position

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

NED frame velocity

Derivative of Euler angles

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$\mathbf{W}$

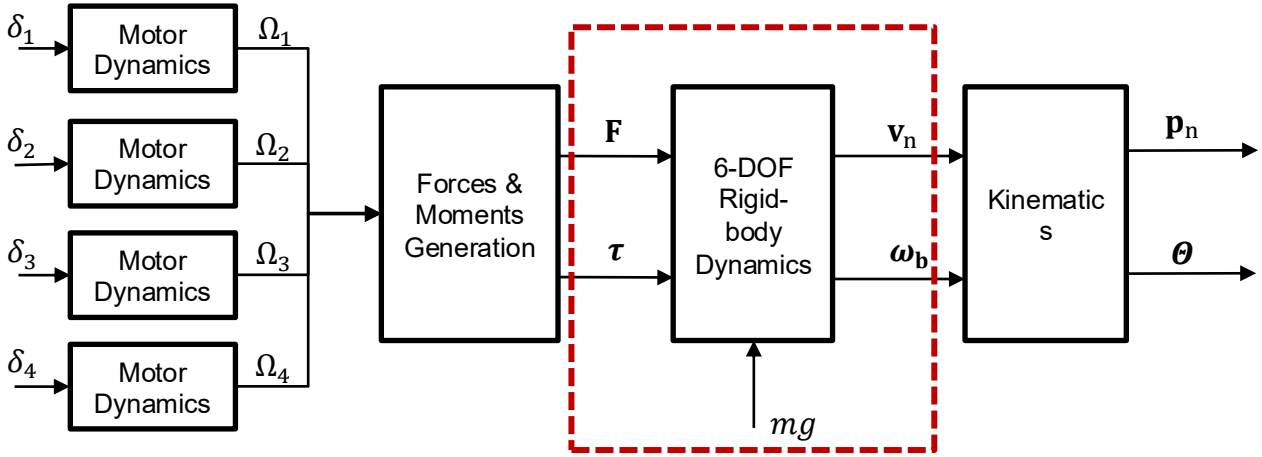
Body-frame angular velocity

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Mathematics Modeling

Rigid-body Dynamics:

Studies the **movement of systems** of interconnected bodies under the action of **external forces**.



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Mathematics Modeling

Rigid-body Dynamics:

Studies the **movement of systems** of interconnected bodies under the action of **external forces**. Based on Newton-Euler formalism describing the translational and rotational dynamics of a rigid body, the dynamic equations can be written into input-output form

Position dynamic model

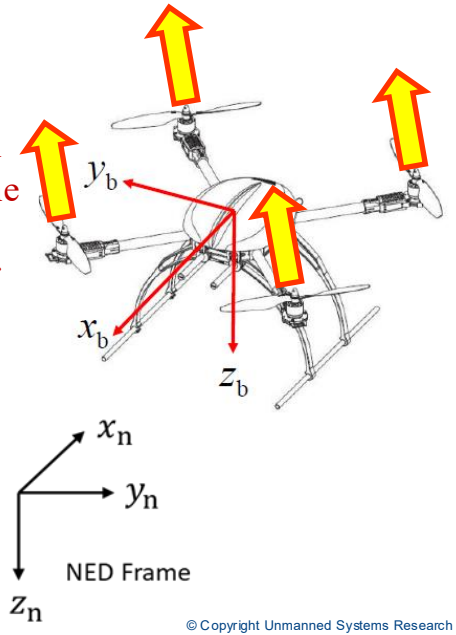
$$\dot{\mathbf{v}}_{\mathrm{n}} = g\mathbf{e_3} + \mathbf{R}_{\mathrm{n/b}}\frac{\mathbf{F}}{m}$$

where $\mathbf{F} = [0, 0, -f]^T = -f\mathbf{e}_3$

$$\dot{\mathbf{v}}_{\mathrm{n}} = g\mathbf{e_3} - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\mathbf{R}_{\mathrm{n/b}} = \begin{bmatrix} c_\theta\, c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta\, s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}$$

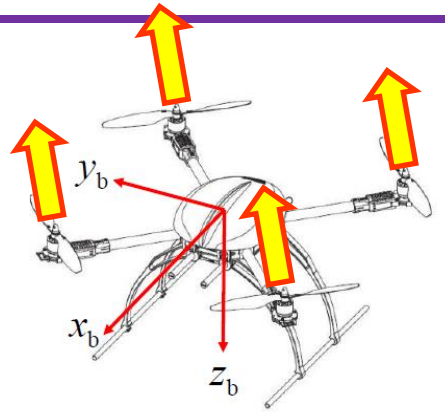
Assumption 4. The multirotor is only under gravity and propeller thrust. Furthermore, the gravity points along the positive direction of the z-axis of NED frame and the propeller thrust points along the negative direction of the z-axis of Body frame.



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Mathematics Modeling

Rigid-body Dynamics:

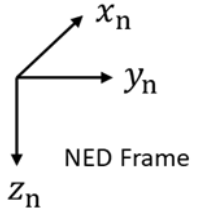


Attitude dynamic model

$$\mathbf{J} \cdot \dot{\boldsymbol{\omega}}_{\mathrm{b}} = -\boldsymbol{\omega}_{\mathrm{b}} \times (\mathbf{J} \cdot \boldsymbol{\omega}_{\mathrm{b}}) + \boldsymbol{G}_{\mathrm{a}} + \boldsymbol{\tau}$$

where $\boldsymbol{\tau} \triangleq [\tau_x, \tau_y, \tau_z] \in \mathbb{R}^3$ represents the moments/torques generated by the propellers in the body axes; $\mathbf{J} \in \mathbb{R}^{3\times3}$ represents the multirotor moments of inertia. G_a denotes gyroscopic torque from the spinning rotors interacting with the body's angular velocity.

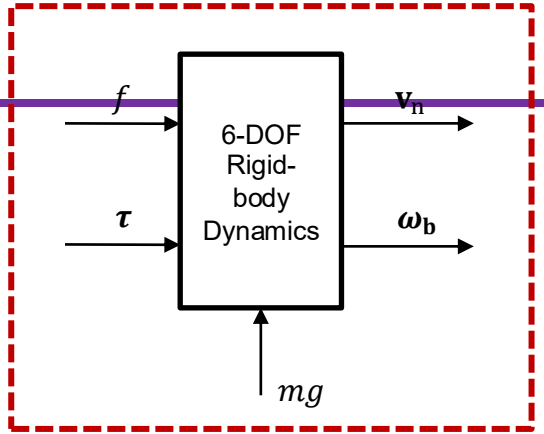
$$\mathbf{J} = \begin{bmatrix} J_x & 0 & 0 \\ 0 & J_y & 0 \\ 0 & 0 & J_z \end{bmatrix}$$



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Mathematics Modeling



Rigid-body Dynamics

Summary of rigid-body dynamics

$$\dot{\mathbf{v}}_\mathrm{n} = g\mathbf{e_3} - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\mathbf{J}\cdot\dot{\boldsymbol{\omega}}_\mathrm{b} = -\boldsymbol{\omega}_\mathrm{b}\times(\mathbf{J}\cdot\boldsymbol{\omega}_\mathrm{b}) + \boldsymbol{G}_\mathrm{a} + \boldsymbol{\tau}$$

NED frame acceleration

$$\begin{bmatrix}\dot{v}_x\\ \dot{v}_y\\ \dot{v}_z\end{bmatrix} = \begin{bmatrix}0\\0\\g\end{bmatrix} - \begin{bmatrix}c_\phi s_\theta c_\psi + s_\phi s_\psi\\ c_\phi s_\theta s_\psi - s_\phi c_\psi\\ c_\phi c_\theta\end{bmatrix}\frac{f}{m}$$

Body frame thrust

Multirotor moments of inertia

Body-frame angular acceleration

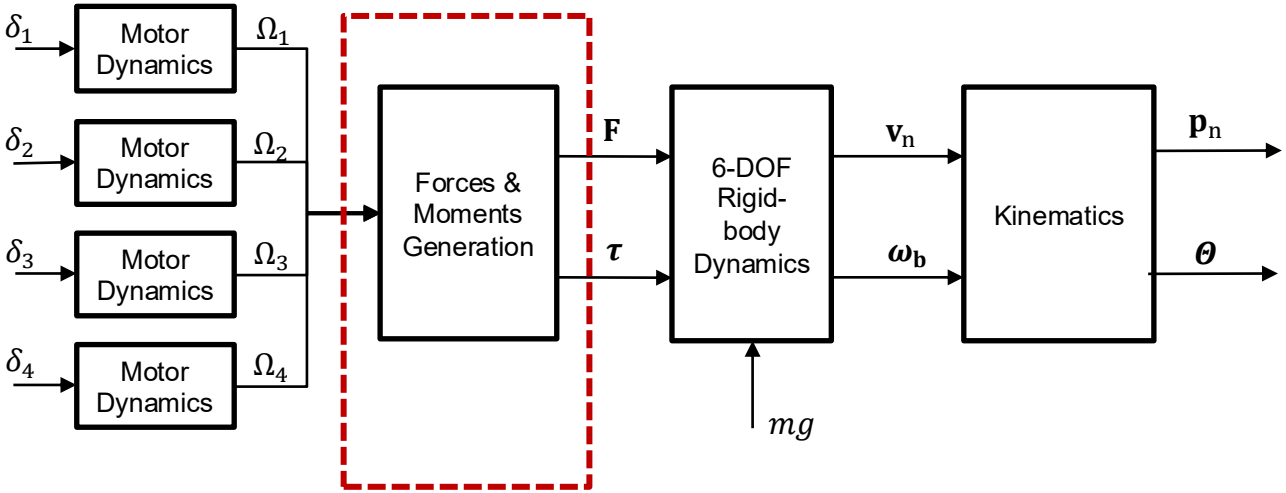
$$\mathbf{J}\cdot\begin{bmatrix}\dot{\omega}_x\\ \dot{\omega}_y\\ \dot{\omega}_z\end{bmatrix} = -\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix}\times\left(\mathbf{J}\cdot\begin{bmatrix}\omega_x\\ \omega_y\\ \omega_z\end{bmatrix}\right) + \boldsymbol{G}_\mathrm{a} + \begin{bmatrix}\tau_x\\ \tau_y\\ \tau_z\end{bmatrix}$$

Body frame resultant torque

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Mathematics Modelling

Forces/Moments Generation



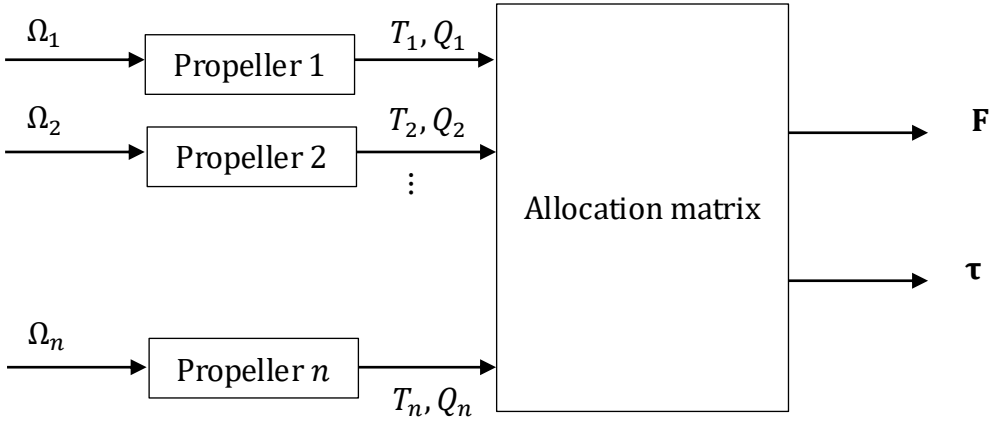
Q2. How should the rotors be arranged?

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Mathematics Modelling

Forces/Moments Generation

- 6 axes force/moment component generated by all propellers

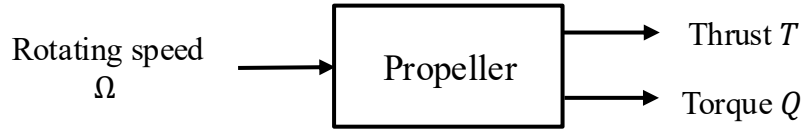


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Mathematics Modelling

Forces/Moment Generation

- Rotating propeller generate both thrust and torque



Aerodynamic thrust coefficient

Diameter of propeller

$$T = \frac{1}{4\pi^2} C_T \rho D^4 \Omega^2$$

$$T = k_T \Omega^2$$

$$Q = \frac{1}{4\pi^2} C_Q \rho D^5 \Omega^2$$

$$Q = k_Q \Omega^2$$

Aerodynamic torque coefficient

Constants

Density of air

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Mathematics Modelling

Force/Moment Generation

- General case

$$F = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} + R_{\mathrm{b/n}} \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}$$

Why rotate?

Gravitational force

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ f_z \end{bmatrix} + mg \begin{bmatrix} -\sin\theta \\ \sin\phi\cos\theta \\ \cos\phi\cos\theta \end{bmatrix}$$

Why '0'?

$= f$ (To simplify the representation)

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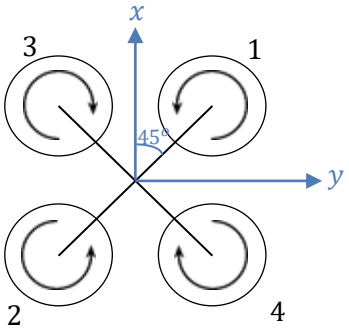
Mathematics Modelling

Force/Moment Generation

- Quad-copter (**X4**)

What does X4 stand for?

- $f, \tau_x, \tau_y, \tau_z$ are **body-frame** force and moment components



$$f = -(T_1 + T_2 + T_3 + T_4)$$

$$\tau_x = \frac{\sqrt{2}l}{2}(-T_1 + T_2 + T_3 - T_4)$$

$$\tau_y = \frac{\sqrt{2}l}{2}(T_1 - T_2 + T_3 - T_4)$$

$$\tau_z = Q_1 + Q_2 - Q_3 - Q_4$$

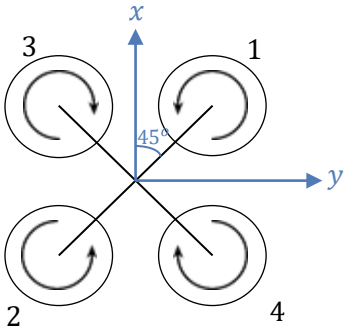
$$T = k_T \Omega^2$$
$$Q = k_Q \Omega^2$$

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Mathematics Modelling

Force/Moment Generation

- Quad-copter (X4)
- $f, \tau_x, \tau_y, \tau_z$ are **body-frame** force and moment components



$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} -k_T & -k_T & -k_T & -k_T \\ -\frac{\sqrt{2}}{2} lk_T & \frac{\sqrt{2}}{2} lk_T & \frac{\sqrt{2}}{2} lk_T & -\frac{\sqrt{2}}{2} lk_T \\ \frac{\sqrt{2}}{2} lk_T & -\frac{\sqrt{2}}{2} lk_T & \frac{\sqrt{2}}{2} lk_T & -\frac{\sqrt{2}}{2} lk_T \\ k_Q & k_Q & -k_Q & -k_Q \end{bmatrix}}_{} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

Unique solution: **Invertible** ↔ Full Rank

Q2. How should the rotors be arranged?

Ans: This matrix must be invertible, so that we can have one solution for each force and moment we desired

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Mathematics Modelling

Force/Moment Generation

- Quad-copter (X4)
- $f, \tau_x, \tau_y, \tau_z$ are **body-frame** force and moment components

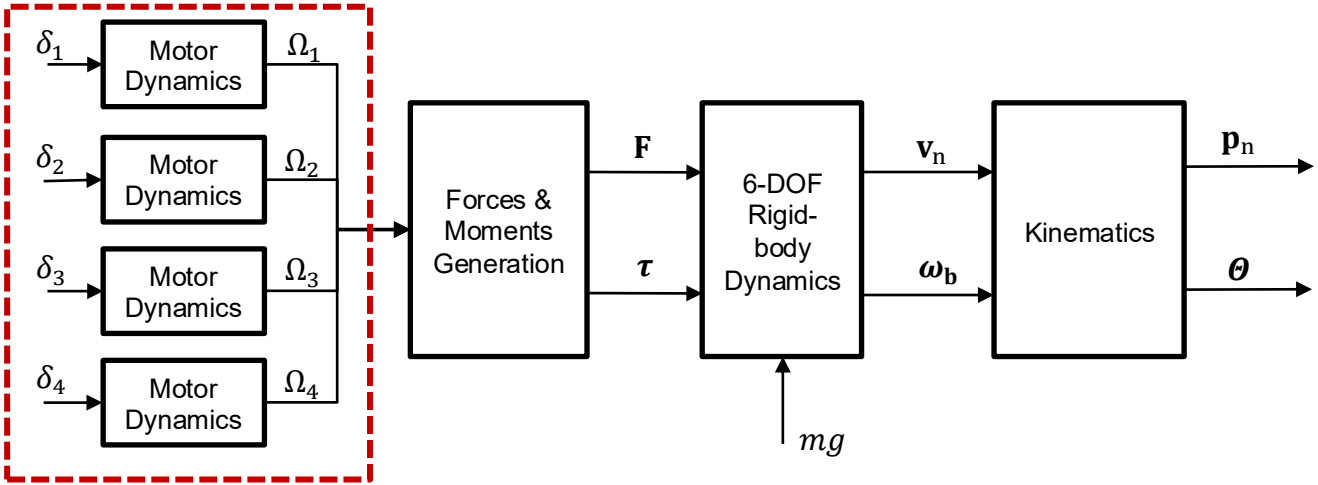
$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} -k_T & -k_T & -k_T & -k_T \\ -\frac{\sqrt{2}}{2}lk_T & \frac{\sqrt{2}}{2}lk_T & \frac{\sqrt{2}}{2}lk_T & -\frac{\sqrt{2}}{2}lk_T \\ \frac{\sqrt{2}}{2}lk_T & \frac{\sqrt{2}}{2}lk_T & -\frac{\sqrt{2}}{2}lk_T & -\frac{\sqrt{2}}{2}lk_T \\ k_Q & k_Q & -k_Q & -k_Q \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -1 & -1 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 1 & 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} k_T & 0 & 0 & 0 \\ 0 & lk_T & 0 & 0 \\ 0 & 0 & lk_T & 0 \\ 0 & 0 & 0 & k_Q \end{bmatrix} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix} = \mathbf{M} \cdot \mathbf{\Gamma} \begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

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Mathematics Modelling

Motor Dynamics



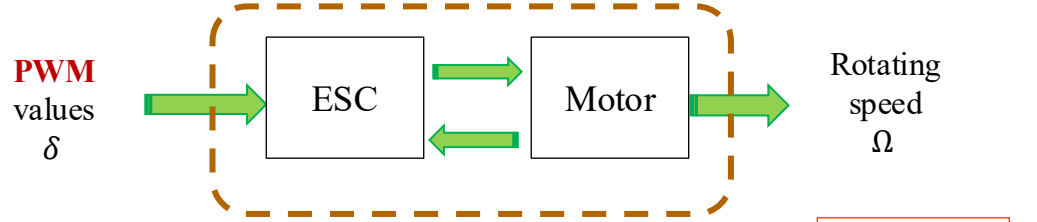
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Mathematics Modelling

Motor Dynamics

Review question: how many ESCs required for X4?

- An Electronics Speed Controller (**ESC**) to control speed of each motor
- Generally, treat ESC-Motor pair as a black-box to be identified



$$\Omega = \frac{1}{T_m s + 1}(K_m \delta + \Omega^*)$$

Motor trim value

Motor time constant

Motor steady state gain

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Mathematics Modelling

Summary of the Mathematics Model of a Quadrotor UAV

Kinematics:

$$\dot{\mathbf{p}}_{\mathrm{n}} = \mathbf{v}_{\mathrm{n}}$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}$$

Dynamics:

$$\dot{\mathbf{v}}_{\mathrm{n}} = g\mathbf{e_3} - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\mathbf{J}\cdot\dot{\boldsymbol{\omega}}_{\mathrm{b}} = -\boldsymbol{\omega}_{\mathrm{b}}\times(\mathbf{J}\cdot\boldsymbol{\omega}_{\mathrm{b}}) + \boldsymbol{G}_{\mathrm{a}} + \boldsymbol{\tau}$$

Motor Dynamics:

$$\Omega = \frac{1}{T_m s + 1}(K_m\delta + \Omega^*)$$

Forces & Moments Generation

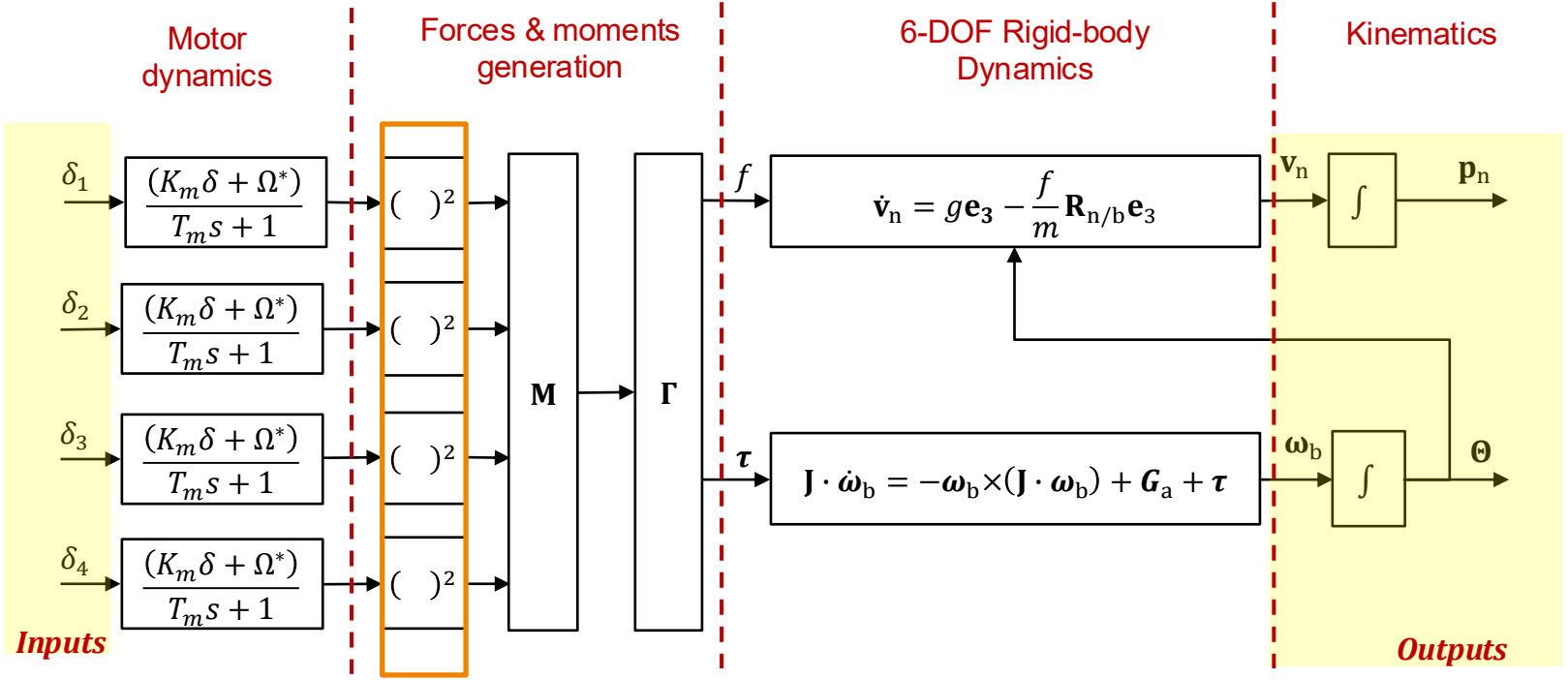
$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \mathbf{M}\cdot\boldsymbol{\Gamma}\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

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Mathematics Modelling

Illustration of the Model for a Quadrotor UAV

How to **identify parameters** in the model?



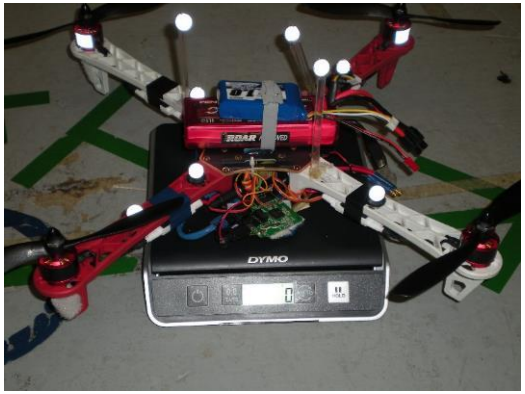
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Parameter Identification

Direct Measurement

- Can be easily measured by ruler/scale

m - Mass



l – Arm length (motor shalf to CG)



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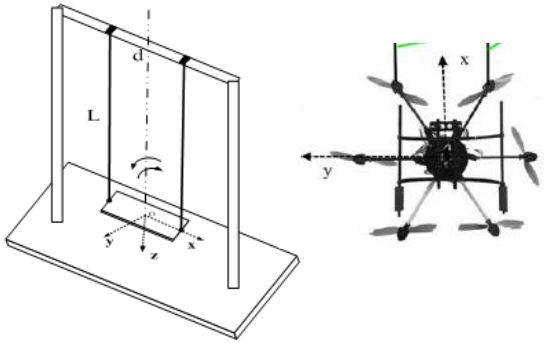
Parameters Identification

Test Bench Experiment (Without External Sensor)

- Can be easily estimated with simple experiment results

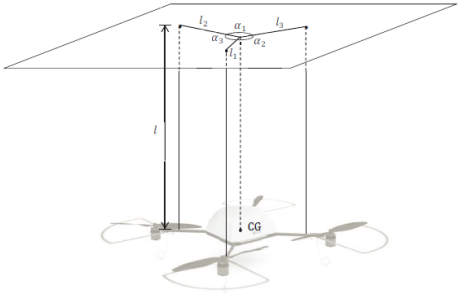
J – Moment of Inertial

Bifilar Pendulum Method



$$J_{zz} = \frac{m_0 g d^2}{16\pi^2 L} T_0^2$$

Trifilar Pendulum Method



$$J_z = \frac{mgl_1 l_2 l_3 t_l^2}{4\pi^2 l} \cdot \frac{l_1 \sin\alpha_1 + l_2 \sin\alpha_2 + l_3 \sin\alpha_3}{l_2 l_3 \sin\alpha_1 + l_1 l_3 \sin\alpha_2 + l_1 l_2 \sin\alpha_3}.$$



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Parameters Identification

Test Bench Experiment (Without External Sensor)

- Can be easily estimated with simple experiment results

J – Moment of Inertial

Bifilar Pendulum Method



$$J_{zz} = \frac{m_0 g d^2}{16\pi^2 L} T_0^2$$

$J_{zz} = 0.143108$

Trifilar Pendulum Method



$$J_z = \frac{mgl_1 l_2 l_3 t_l^2}{4\pi^2 l} \cdot \frac{l_1 \sin\alpha_1 + l_2 \sin\alpha_2 + l_3 \sin\alpha_3}{l_2 l_3 \sin\alpha_1 + l_1 l_3 \sin\alpha_2 + l_1 l_2 \sin\alpha_3}.$$

$J_{zz} = 0.140443$

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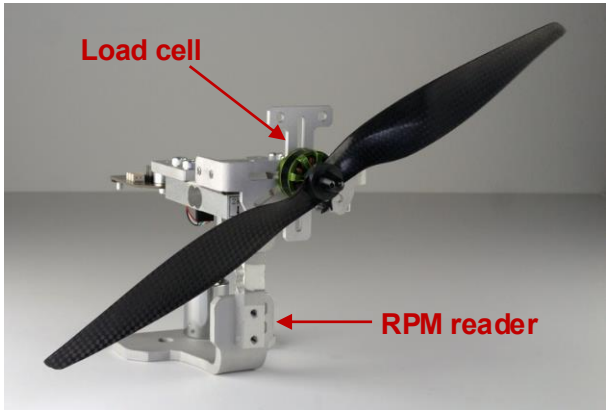
Parameters Identification

Test Bench Experiment (With External Sensors)

- More complicated experimental setup, need help on external sensors such as load cell, RPM reader, etc.

Motor Parameters

$$\Omega = \frac{1}{T_m s + 1}(K_m \delta + \Omega^*)$$



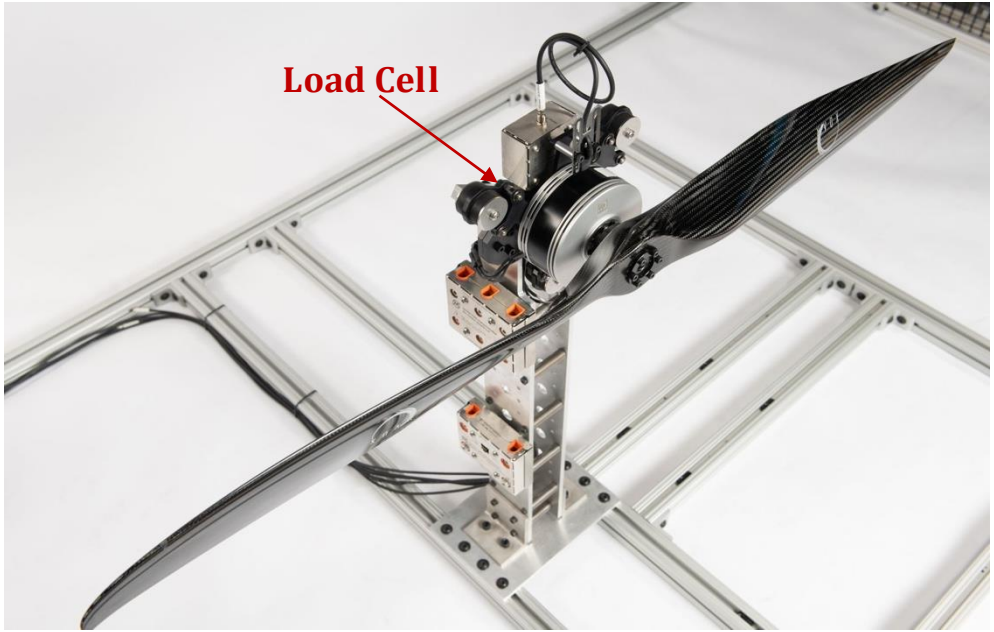
T_m	Step input to motor, time taken to reach 63.2% of steady-state speed.
K_m, Ω^*	Gradually increase input value, record down steady-state speed of each instances. Plot best-fit line to find gradient and offset.

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Parameters Identification

Test Bench Experiment (With External Sensors)

- More complicated experimental setup, need help on external sensors such as load cell, RPM reader, etc.



Propeller Parameters

$$T = k_T \Omega^2$$
$$Q = k_Q \Omega^2$$

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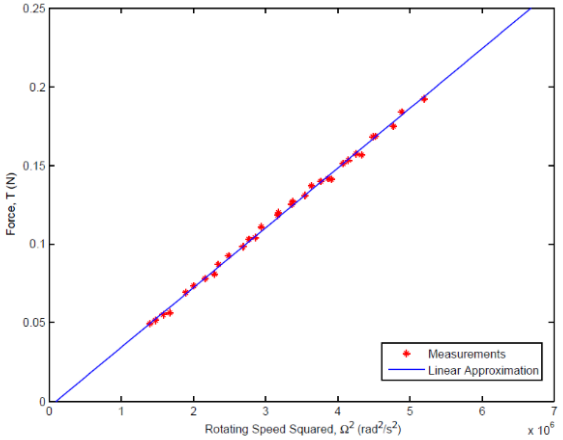
Parameters Identification

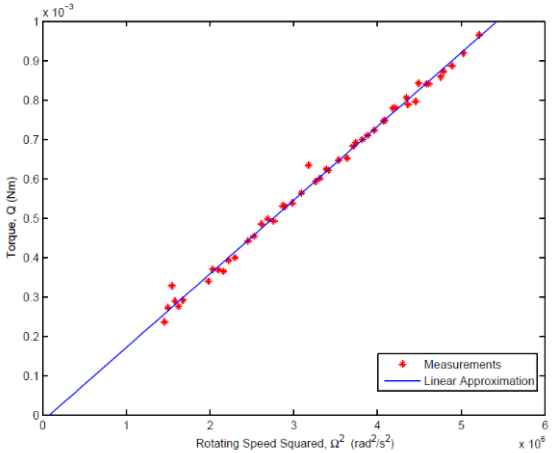
Test Bench Experiment (With External Sensors)

- More complicated experimental setup, need help on external sensors such as load cell, RPM reader, etc.

Propeller Parameters

$$T = k_T \Omega^2$$
$$Q = k_Q \Omega^2$$



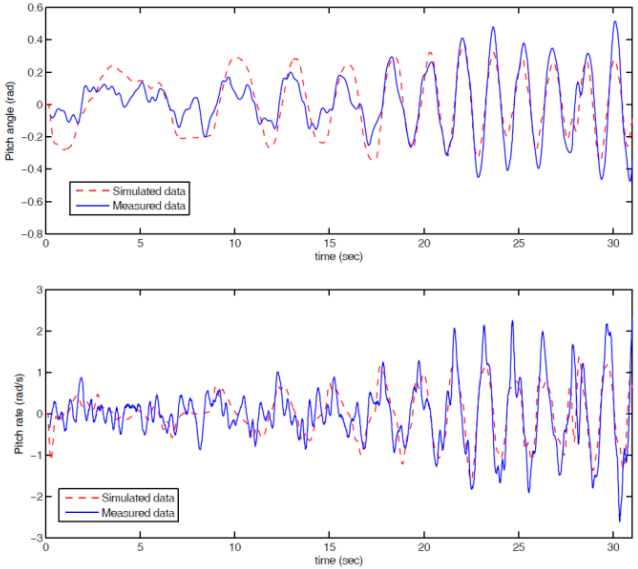


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Model Verification

Time Domain

- Obtain in-flight data for both inputs and outputs of the system
- Run simulated system with mathematical model using the same inputs
- Compare outputs of simulated system and flight data



Chirp input signals!!

Why?

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Summary

Frame Systems

- Commonly split into 2 frames: Navigation (global) frame & Body frame
- Relationship between the 2 frames can be transformed with a rotation matrix

Mathematics Modeling

- System identification or first-principles modeling approaches
- Relates physical aircraft movements to mathematics equations
- Design multi-rotor UAVs based on mathematics equations

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Summary

Parameters Identification

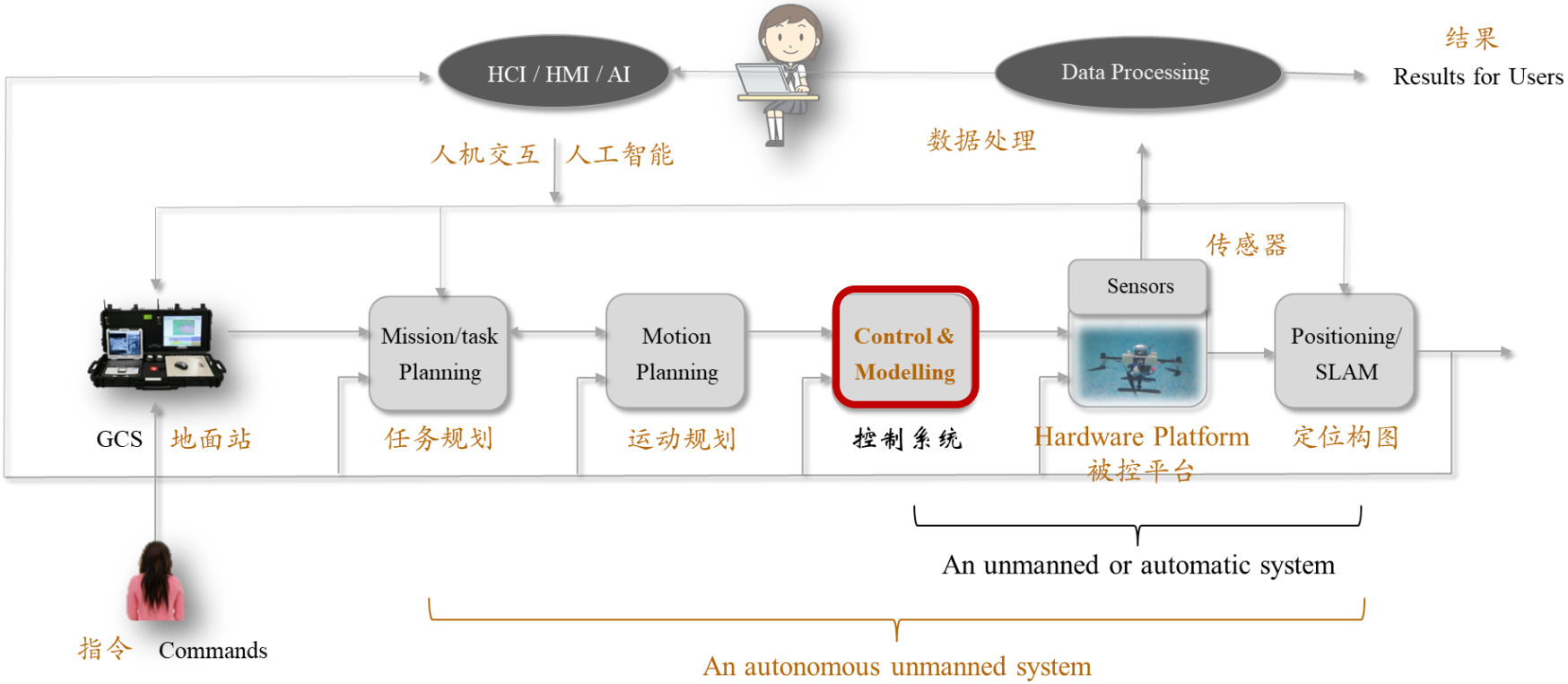
- Identify unknown constant parameter in mathematics model
- Different approach to identify different parameters

Model Verification

- Verify with actual flying by comparing with simulation results

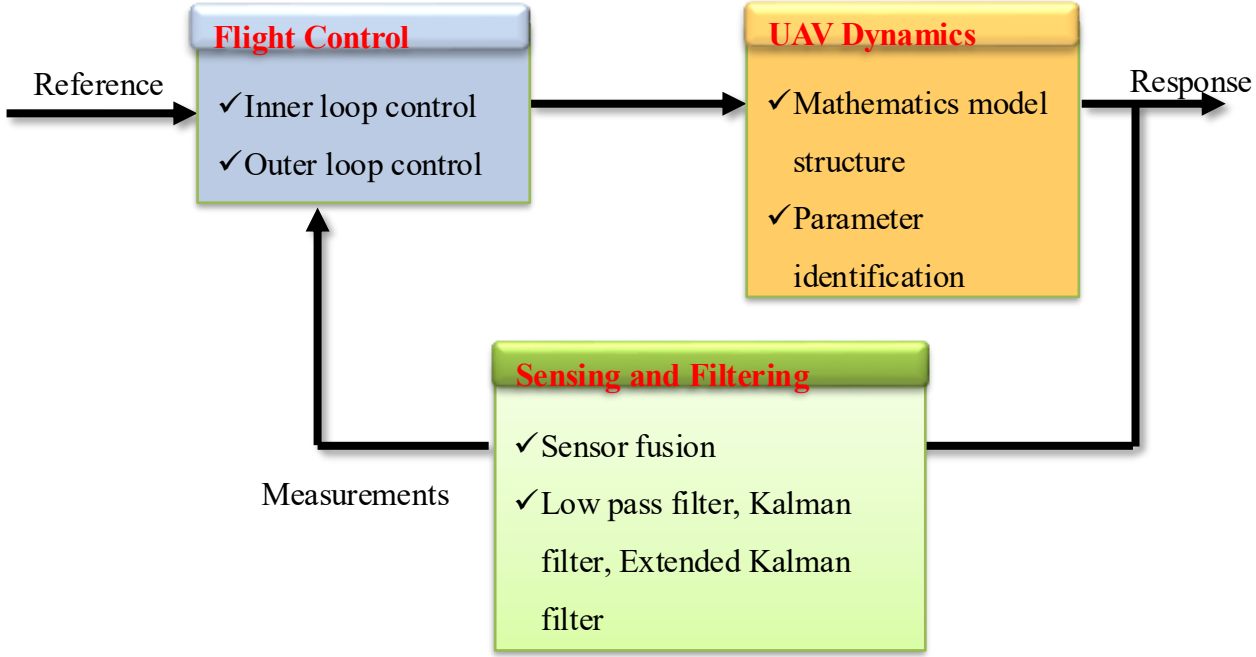
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Flight Control



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Introduction – System Overview



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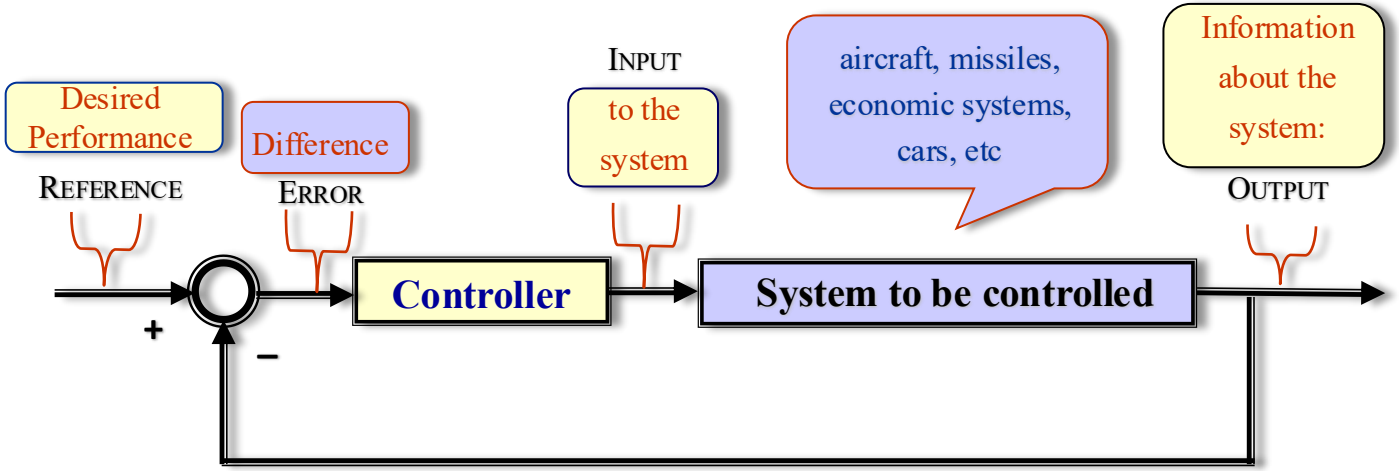


Outline

- Introduction of UAV Flight Control
- Model Simplification
- Flight Control Design
  - Attitude Control (inner loop)
  - Position Control (outer loop)
  - Control Allocation
- Comprehensive Simulation
- Summary

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UAV Control System



Objective: To make the system **Output** and the desired **Reference** as close as possible, i.e., to make the **Error** as small as possible.

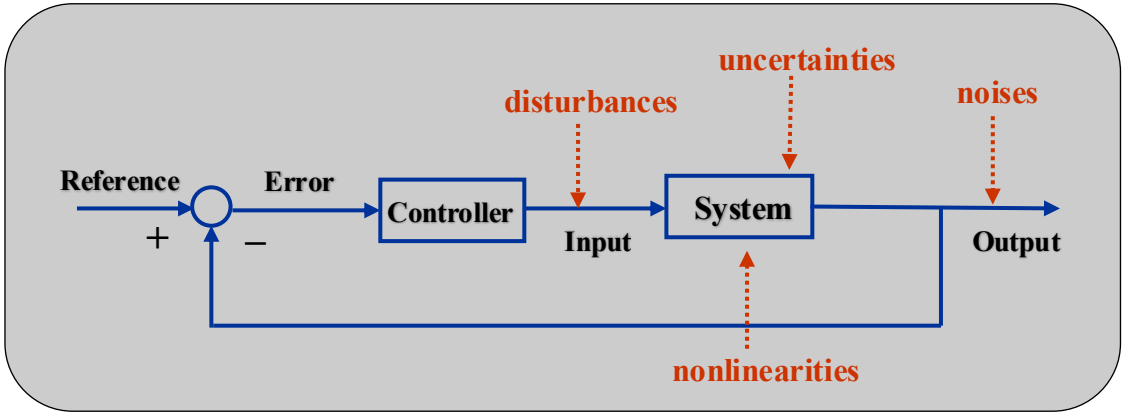
Key Issues: (1) How to describe the system to be controlled? **(Modeling)**

(2) How to design the controller? **(Control)**

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UAV Control System

There are many other factors to be carefully considered when designing a control system for real-life problems. These factors include:



These real-life factors, i.e., disturbances, uncertainties, **nonlinearities**, give birth of many modern control techniques…

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UAV Control System

Control Techniques:

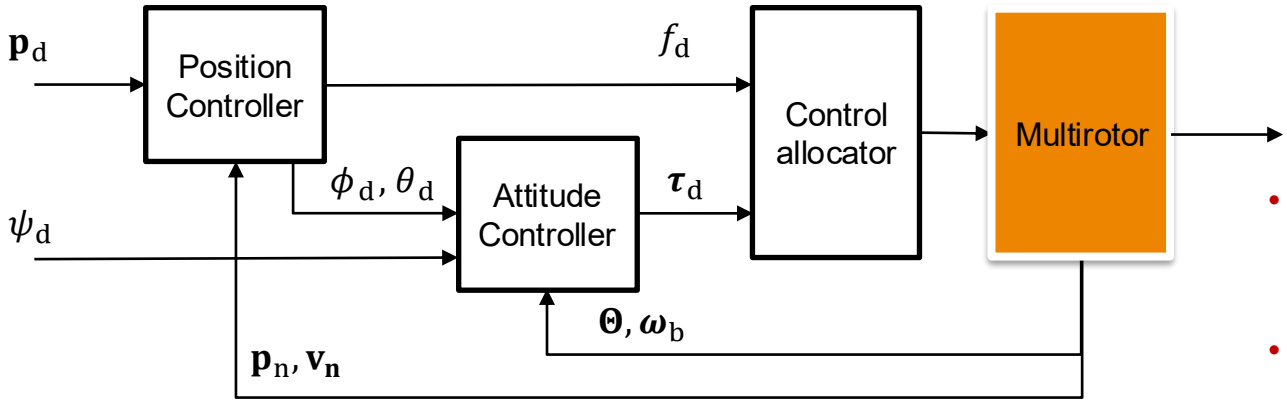
- Classical control: PID control, developed in 1940s and utilized heavily for in industrial processes. Examples: everywhere in life…
- Optimal control: Linear quadratic regulator (LQR), H_2 optimal control, Kalman filter, developed in 1960s to achieve certain optimal performance and boomed by *NASA Apollo Project*.
- Robust control: H_∞ control, developed in 1980s and 90s to handle systems with uncertainties and disturbances and with high performances; *RPT (Robust & Perfect Tracking)*, developed by Chen and Lin (2000), is to design a controller such that the resulting closed-loop system is stable and the controlled output almost perfectly tracks a given reference signal in the presence of any initial conditions and external disturbances
- Nonlinear control: Still on-going research topics, developed to handle nonlinear systems with high performances.
- Intelligent control: Knowledge-based control, adaptive control, neural and fuzzy control… researched heavily in 1990s, developed to handle systems with unknown models. Examples: economic systems, social systems...

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Framework of Flight Control

The flight control of multirotor UAVs includes the following parts:

- Position control: desired position $\mathbf{p}_\mathrm{d} \rightarrow$ desired roll angle ϕ_d, pitch angle θ_d and total thrust f_d
- Attitude control: desired attitude angles $\phi_\mathrm{d}, \theta_\mathrm{d}, \psi_\mathrm{d} \rightarrow$ desired moment $\boldsymbol{\tau}_\mathrm{d}$
- Control allocation: $f_\mathrm{d}, \boldsymbol{\tau}_\mathrm{d} \rightarrow$ desired propeller angular speeds Ω_k, $k = 1,2, \ldots, n$



- Four independent inputs (total thrust f and three-axis moment $\boldsymbol{\tau}$),
- Six outputs (position $\mathbf{p}_\mathrm{n}$ and attitude $\boldsymbol{\Theta}$).

Closed-loop structure of a fight control system for a multirotor UAV

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Outline

- Introduction of UAV Flight Control
- Model Simplification
- Flight Control Design
  - Position Control
  - Attitude Control
  - Control Allocation
- Comprehensive Simulation
- Summary

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Model Simplification

Based on the nonlinear model in previous slides, by ignoring the term of $-\boldsymbol{\omega}_{\mathrm{b}}\times(\mathbf{J}\cdot\boldsymbol{\omega}_{\mathrm{b}})+\boldsymbol{G}_{\mathrm{a}}$, the simplified model is

$$\left.\begin{aligned}\dot{\mathbf{p}}_{\mathrm{n}}&=\mathbf{v}_{\mathrm{n}}\\ \dot{\boldsymbol{\Theta}}&=\mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}\end{aligned}\right\}\Rightarrow$$

$$\dot{\mathbf{v}}_{\mathrm{n}}=g\mathbf{e_3}-\frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\mathbf{J}\cdot\dot{\boldsymbol{\omega}}_{\mathrm{b}}=-\boldsymbol{\omega}_{\mathrm{b}}\times(\mathbf{J}\cdot\boldsymbol{\omega}_{\mathrm{b}})+\boldsymbol{G}_{\mathrm{a}}+\boldsymbol{\tau}$$

$$\ddot{p}_x=-\frac{f}{m}(\sin\psi\sin\phi+\cos\psi\sin\theta\cos\phi)$$

$$\ddot{p}_y=-\frac{f}{m}(-\cos\psi\sin\phi+\cos\phi\sin\theta\sin\psi)$$

$$\ddot{p}_z=g-\frac{f}{m}\cos\theta\cos\phi$$

where $\mathbf{p}_{\mathrm{n}}=[p_x,p_y,p_z]^{\mathrm{T}}$, the specific definitions are referred to previous slides for mathematical modelling.

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Model Simplification

Some assumptions are made that $\sin\phi \approx \phi, \cos\phi \approx 1, \sin\theta \approx \theta, \cos\theta \approx 1, f \approx mg$. Then,

$$\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3 \approx \begin{bmatrix} \theta\cos\psi + \phi\sin\psi \\ \theta\sin\psi - \phi\cos\psi \\ 1 \end{bmatrix}$$

Three linear models are obtained:

(1) Horizontal position channel model

$$\begin{aligned} \dot{\mathbf{p}}_{\mathrm{n}} &= \mathbf{v}_{\mathrm{n}} \\ \dot{\mathbf{v}}_{\mathrm{n}} &= g\mathbf{e}_3 - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3 \end{aligned} \Rightarrow \begin{aligned} \ddot{p}_x &= -\frac{f}{m}(\sin\psi\sin\phi + \cos\psi\sin\theta\cos\phi) \\ \ddot{p}_y &= -\frac{f}{m}(-\cos\psi\sin\phi + \cos\phi\sin\theta\sin\psi) \end{aligned} \Rightarrow \begin{aligned} \ddot{p}_x &= -g(\sin\psi\,\phi + \cos\psi\,\theta) \\ \ddot{p}_y &= -g(-\cos\psi\,\phi + \sin\psi\,\theta) \end{aligned}$$

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$
$$\begin{bmatrix} \dot{v}_x \\ \dot{v}_y \end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix} \phi \\ \theta \end{bmatrix}$$

where $\mathbf{A}_\psi = \begin{bmatrix} \sin\psi & \cos\psi \\ -\cos\psi & \sin\psi \end{bmatrix}$

Since $-g\boldsymbol{A}_\Psi$ is known, the term $-g\boldsymbol{A}_\Psi\begin{bmatrix}\phi\\\theta\end{bmatrix}$ can be viewed as the input. Thus, the horizontal position channel model is linear.

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Model Simplification

(2) Altitude channel model

$$\dot{\mathbf{p}}_{\mathrm{n}} = \mathbf{v}_{\mathrm{n}}$$
$$\dot{\mathbf{v}}_{\mathrm{n}} = g\mathbf{e_3} - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\Rightarrow \quad \ddot{p}_z = g - \frac{f}{m}\cos(\theta)\cos(\phi)$$

Linear model

$$\Rightarrow \quad \boxed{\ddot{p}_z = g - \frac{f}{m}}$$

(3) Attitude model

$$\dot{\boldsymbol{\Theta}} = \mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}$$
$$\mathbf{J}\dot{\boldsymbol{\omega}}_{\mathrm{b}} = \boldsymbol{\tau}$$

Small-angle assumption

$$\boldsymbol{W} \approx \boldsymbol{I}$$

Linear model

$$\boxed{\begin{aligned}\dot{\boldsymbol{\Theta}} &= \boldsymbol{\omega}_{\mathrm{b}} \\ \mathbf{J}\dot{\boldsymbol{\omega}}_{\mathrm{b}} &= \boldsymbol{\tau}\end{aligned}}$$

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Model Simplification

Simplify the motor dynamics

$$\Omega = \frac{1}{T_m s + 1}(K_m \delta + \Omega^*)$$

Since the time constant of the motor is very small (in the range of ms), in most applications for controller design, we can consider the steady state condition only, and set trim value $\Omega^* = 0$. Therefore, we may have the following relationship:

$$\Omega_i = K_m \delta_i$$

Additionally, in the hovering state, we can have

$$\Omega_{\text{hover}} = K_m \delta_{\text{hover}}$$

To simplify the formula, we may use Ω_0 and δ_0 to denote the rotor speed and motor input in the hovering condition in the lecture notes.

$$\Omega_0 = K_m \delta_0$$

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Model Simplification

Linearization of Forces & Moments Generation

We linearize the square of the rotation speed of the rotor. A common approach is to perform a first-order Taylor expansion of Ω_i^2 around a working point, which is typically the hover speed Ω_0:

$$\Omega_i^2 \approx \Omega_0^2 + 2\Omega_0(\Omega_i - \Omega_0) = \Omega_0^2\left(1 + 2\frac{\Omega_i - \Omega_0}{\Omega_0}\right) = \Omega_0^2(1 + 2\Delta\bar{\Omega}_i)$$

This leads to:

$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & -1 & -1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}}_{\mathbf{M}} \underbrace{\begin{bmatrix} k_T & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2}lk_T & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2}lk_T & 0 \\ 0 & 0 & 0 & k_Q \end{bmatrix}}_{\mathbf{\Gamma}} \begin{bmatrix} \Omega_0^2(1 + 2\Delta\bar{\Omega}_1) \\ \Omega_0^2(1 + 2\Delta\bar{\Omega}_2) \\ \Omega_0^2(1 + 2\Delta\bar{\Omega}_3) \\ \Omega_0^2(1 + 2\Delta\bar{\Omega}_4) \end{bmatrix} = \mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \begin{bmatrix} 1 + 2\Delta\bar{\Omega}_1 \\ 1 + 2\Delta\bar{\Omega}_2 \\ 1 + 2\Delta\bar{\Omega}_3 \\ 1 + 2\Delta\bar{\Omega}_4 \end{bmatrix}$$

The wrench deviations depend linearly on the normalized rotor deviations $\Delta\bar{\Omega}_i$:

$$\Delta\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} - \begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}_0 = \mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \begin{bmatrix} 1 + 2\Delta\bar{\Omega}_1 \\ 1 + 2\Delta\bar{\Omega}_2 \\ 1 + 2\Delta\bar{\Omega}_3 \\ 1 + 2\Delta\bar{\Omega}_4 \end{bmatrix} - \begin{bmatrix} -mg \\ 0 \\ 0 \\ 0 \end{bmatrix}_0 = \mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \begin{bmatrix} 2\Delta\bar{\Omega}_1 \\ 2\Delta\bar{\Omega}_2 \\ 2\Delta\bar{\Omega}_3 \\ 2\Delta\bar{\Omega}_4 \end{bmatrix}$$

Hovering condition

$$\mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -mg \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

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Model Simplification

Kinematics:

$$\dot{\mathbf{p}}_{\mathrm{n}} = \mathbf{v}_{\mathrm{n}}$$

$$\dot{\boldsymbol{\Theta}} = \mathbf{W}\,\boldsymbol{\omega}_{\mathrm{b}}$$

Dynamics:

$$\dot{\mathbf{v}}_{\mathrm{n}} = g\mathbf{e_3} - \frac{f}{m}\mathbf{R}_{\mathrm{n/b}}\mathbf{e}_3$$

$$\mathbf{J}\cdot\dot{\boldsymbol{\omega}}_{\mathrm{b}} = -\boldsymbol{\omega}_{\mathrm{b}}\times(\mathbf{J}\cdot\boldsymbol{\omega}_{\mathrm{b}}) + \boldsymbol{G}_{\mathrm{a}} + \boldsymbol{\tau}$$

Motor Dynamics:

$$\Omega = \frac{1}{T_m s + 1}(K_m\delta + \bar{\Omega})$$

Forces & Moments Generation

$$\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \mathbf{M}\cdot\boldsymbol{\Gamma}\begin{bmatrix} \Omega_1^2 \\ \Omega_2^2 \\ \Omega_3^2 \\ \Omega_4^2 \end{bmatrix}$$

1) Horizontal position channel model

$$\begin{bmatrix} \dot{p}_x \\ \dot{p}_y \end{bmatrix} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

$$\begin{bmatrix} \dot{v}_x \\ \dot{v}_y \end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix} \phi \\ \theta \end{bmatrix}$$

2) Altitude channel model

$$\ddot{p}_z = g - \frac{f}{m}$$

3) Attitude channel model

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\omega}_{\mathrm{b}}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}}_{\mathrm{b}} = \boldsymbol{\tau}$$

Each channel is a linear model!

4) Motor

$$\Omega_i = K_m\delta_i$$

5) Forces & Moments Generation

$$\Delta\begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \mathbf{M}\cdot\boldsymbol{\Gamma}\cdot\Omega_0^2\begin{bmatrix} 2\Delta\bar{\Omega}_1 \\ 2\Delta\bar{\Omega}_2 \\ 2\Delta\bar{\Omega}_3 \\ 2\Delta\bar{\Omega}_4 \end{bmatrix}$$

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Model Simplification

Simplified Model for Control Design

Finally, we can derived the classical linear kinematics and dynamics model for control design.

1) Horizontal position channel model

$$\begin{bmatrix}\dot{p}_x\\ \dot{p}_y\end{bmatrix} = \begin{bmatrix}v_x\\ v_y\end{bmatrix}$$

$$\begin{bmatrix}\dot{v}_x\\ \dot{v}_y\end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix}\phi\\ \theta\end{bmatrix}$$

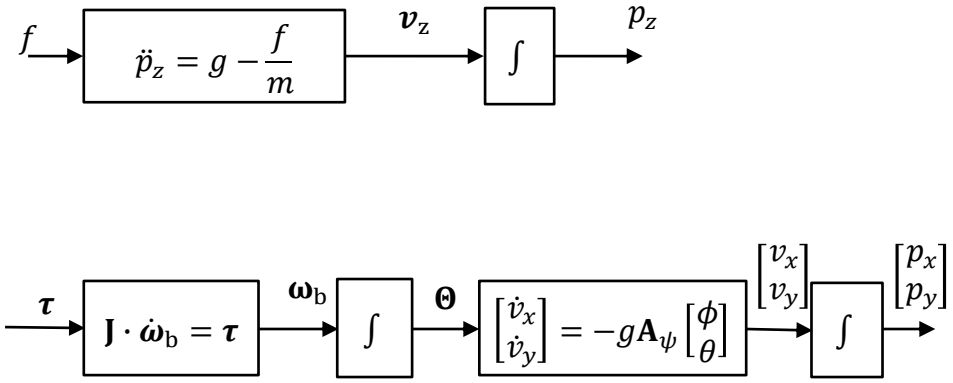
2) Altitude channel model

$$\ddot{p}_z = g - \frac{f}{m}$$

3) Attitude channel model

$$\dot{\boldsymbol{\Theta}} = \boldsymbol{\omega}_\mathrm{b}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}}_\mathrm{b} = \boldsymbol{\tau}$$



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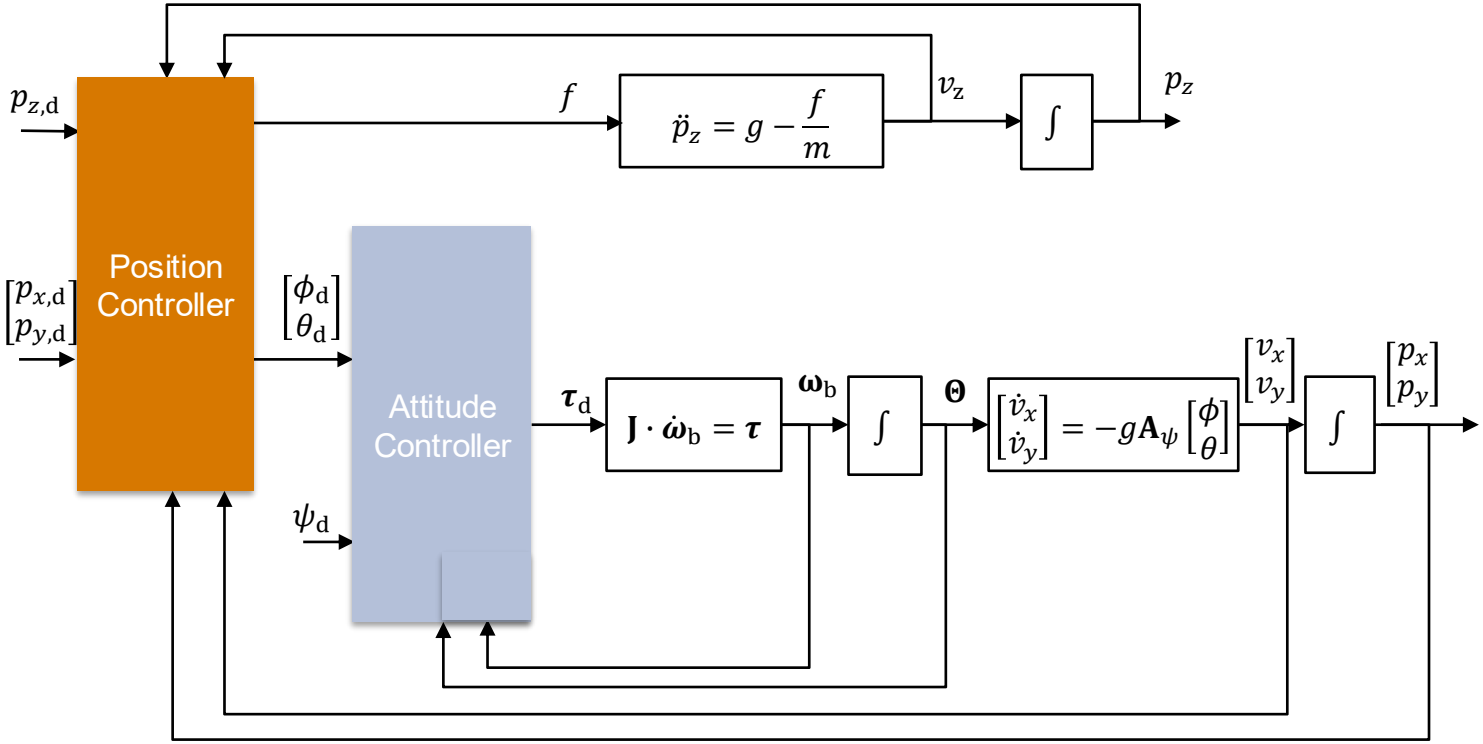
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- Conclusion

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Model Simplification

Framework of Flight Control with Linear Model



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Attitude Control

Basic Concept

Attitude control is the basis of the position control. At present, the commonly-used attitude representations for a rigid body include the Euler angles and rotation matrix.

Comparison of attitude representation

Attitude representations	Advantages	Disadvantages
Euler angles	No redundant parameter; Clear physical meaning	Singularity problem when the pitch angle is 90 degrees; Plenty of transcendental function operation; Gimbal Lock
Rotation matrix	No singularity problem; No transcendental function operation; Applicability for the continuous rotation; Global and unique; Easy to interpolate	Six redundant parameters

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Attitude Control

Euler Angles Based Attitude Control

Objective:

Design a control input $\boldsymbol{\tau}_{\mathrm{d}}$ such that the attitude error

$$\boldsymbol{e}_{\boldsymbol{\Theta}} \triangleq \boldsymbol{\Theta} - \boldsymbol{\Theta}_{\mathrm{d}} \quad \text{satfisfy} \quad \lim_{t\to\infty} \|\mathbf{e}_{\boldsymbol{\Theta}}\| = 0$$

where $\boldsymbol{\Theta}_{\mathrm{d}} = [\phi_{\mathrm{d}}, \theta_{\mathrm{d}}, \psi_{\mathrm{d}}]^T$ is the desired attitude:

- $[\phi_{\mathrm{d}}, \theta_{\mathrm{d}}]^T$ from the position controller
- ψ_{d} from the mission planner

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Attitude Control

Euler Angles Based Attitude Control

Angle Control Loop:

Using $\dot{\boldsymbol{\Theta}} = \boldsymbol{\omega}$, define the desired angular velocity as:

$$\boldsymbol{\omega}_{\mathrm{d}} = -\mathbf{K}_{\Theta}\mathbf{e}_{\Theta}\,, \qquad \mathbf{K}_{\Theta} \in \mathbb{R}^{3\times 3}$$

Angular Rate Control Loop:

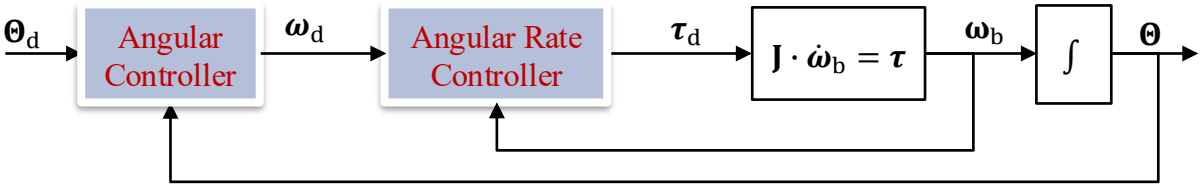
Based on $\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}$ design the moment input as:

$$\boldsymbol{\tau}_{\mathrm{d}} = -\mathbf{K}_{\omega\mathrm{p}}\mathbf{e}_{\omega} - \mathbf{K}_{\omega\mathrm{i}}\int \mathbf{e}_{\omega}\, dt - \mathbf{K}_{\omega\mathrm{d}}\dot{\mathbf{e}}_{\omega}$$

where $\mathbf{e}_{\omega} \triangleq \boldsymbol{\omega} - \boldsymbol{\omega}_{\mathrm{d}}$, and $\mathbf{K}_{\omega\mathrm{p}}, \mathbf{K}_{\omega\mathrm{i}}, \mathbf{K}_{\omega\mathrm{d}} \in \mathbb{R}^{3\times 3}$

What control method/technique?

How to add saturation?



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Outline

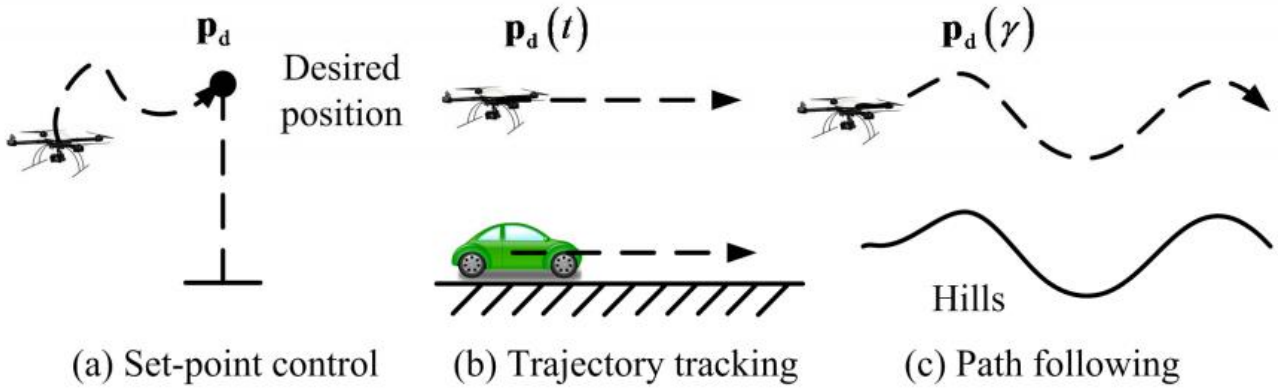
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Position Control

Classification

According to the given position $\mathbf{p}_d$, the position control is divided into three types: the set-point control, trajectory tracking and path following.



Three types of position control

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Position Control

Classification

1) Set-Point Control. The desired position $\mathbf{p}_\mathrm{d} \in \mathbb{R}^3$ is a constant point. The aim of this part is to design a controller to make $\|\mathbf{p}(t) - \mathbf{p}_\mathrm{d}\| \to 0$.

2) Trajectory Tracking. The desired trajectory $\mathbf{p}_\mathrm{d}(t)$ is a time-dependent trajectory. The aim of this part is to design a controller to make $\|\mathbf{p}(t) - \mathbf{p}_\mathrm{d}(t)\| \to 0$.

3) Path Following [1]. The desired path $\mathbf{p}_\mathrm{d}(\gamma) \in \mathbb{R}^3$ is a path directly determined by the parameter γ rather than the time t. It aims to design a controller to make $\|\mathbf{p}(t) - \mathbf{p}_\mathrm{d}(t)\| \to 0$.

$P_d(\gamma\ (t)\ )$

[1] Hespanha J P. Trajectory-tracking and path-following of underactuated autonomous vehicles with parametric modeling uncertainty. Automatic Control, IEEE Transactions on, 2007, 52(8): 1362–1379.

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Position Control

Classification

Desired attitude is the output of position controllers. According to the attitude representation, the position control is divided into two types:

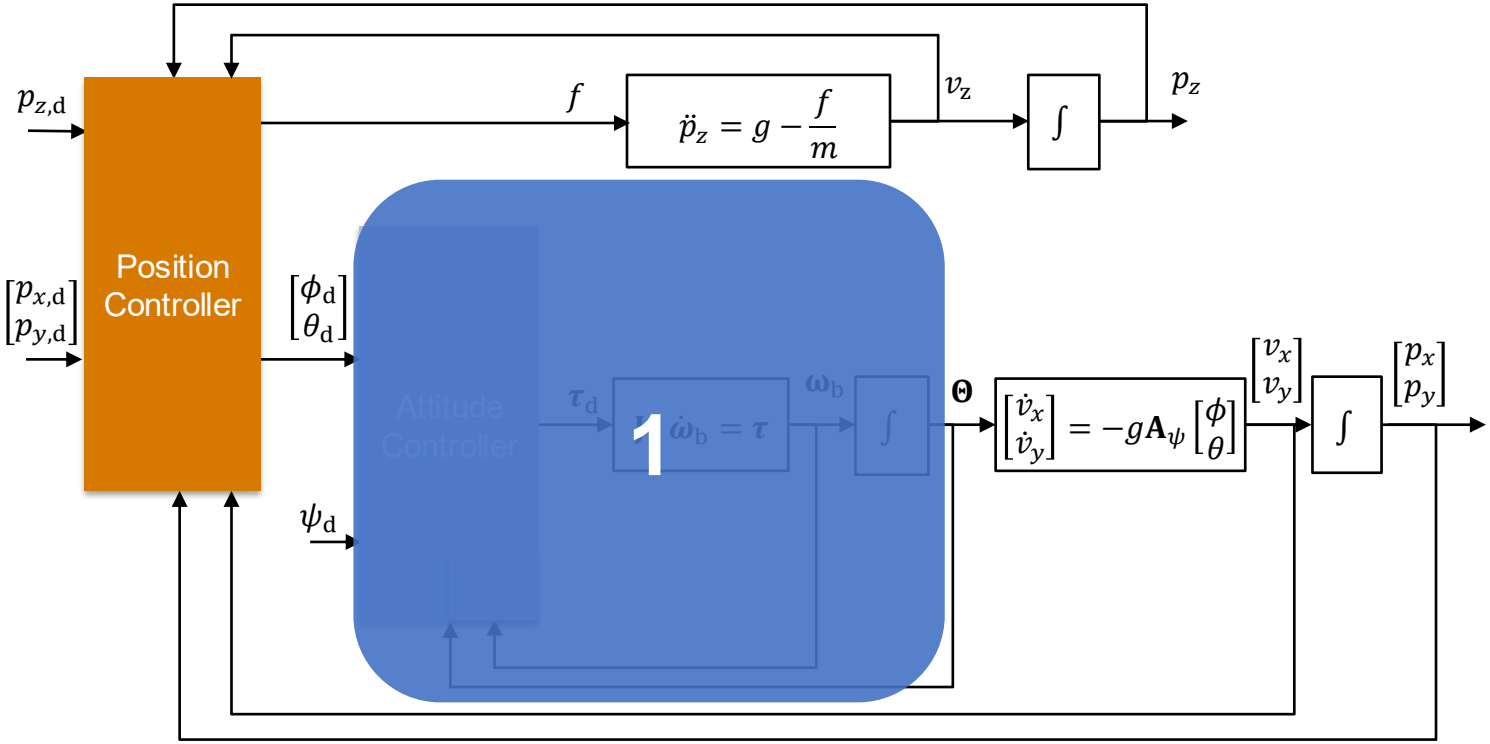
(1) Euler angles as output. The controller, which is designed according to the linear systems, generates the desired Euler angles and thrust $\phi_{\mathrm{d}}, \theta_{\mathrm{d}}, f_{\mathrm{d}}$.

(2) Rotation matrix as output. The controller, which is designed according to the nonlinear coupled systems, generates the desired rotation matrix and thrust $\boldsymbol{R}_{\mathrm{d}}, f_{\mathrm{d}}$.

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Position Control

Simplified Model with Control



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Position and Velocity Control

Outer-loop dynamics

1) Horizontal position channel model

$$\begin{bmatrix}\dot{p}_x\\ \dot{p}_y\end{bmatrix} = \begin{bmatrix}v_x\\ v_y\end{bmatrix}$$

$$\begin{bmatrix}\dot{v}_x\\ \dot{v}_y\end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix}\phi\\ \theta\end{bmatrix}$$

2) Altitude channel model

$$\ddot{p}_z = g - \frac{f}{m}$$

$$\dot{\mathbf{p}} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\mathbf{v}$$

$$\dot{\mathbf{v}} = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\mathbf{a}$$

where

$$\begin{bmatrix}a_x\\ a_y\end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix}\phi\\ \theta\end{bmatrix}$$

$$a_z = g - \frac{f}{m}$$

Objective

The position control objective is to design a controller that drives the quadrotor from its current position $\mathbf{p}(t) \in \mathbb{R}^3$ to a desired position $\mathbf{p_d}(t) \in \mathbb{R}^3$, ensuring that the position error $\mathbf{e}_p(t) = \mathbf{p}(t) - \mathbf{p}_\mathrm{d}(t)$ asymptotically converges to zero:

$$\lim_{t\to\infty} \|\mathbf{e}_p(t)\| = 0$$

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Position and Velocity Control

Control Structure

We design a second-order PID controller with a feedforward velocity and acceleration terms. The desired total acceleration a_d is composed of:

- Proportional feedback on position error
- Derivative feedback on velocity error
- Feedforward desired acceleration

$$\boldsymbol{a}_{\mathrm{d}} = \begin{bmatrix} a_{x,\mathrm{d}} \\ a_{y,\mathrm{d}} \\ a_{z,\mathrm{d}} \end{bmatrix} = \mathbf{F}_{\mathrm{p}}(\mathbf{p}_{\mathrm{ref}} - \mathbf{p}) + \mathbf{F}_{\mathrm{I}} \int (\mathbf{p}_{\mathrm{ref}} - \mathbf{p}) dt + \mathbf{F}_{\mathrm{v}}(\mathbf{v}_{\mathrm{ref}} - \mathbf{v}) + \boldsymbol{a}_{\mathrm{ref}}$$

where:

- $\mathbf{F}_{\mathrm{p}}, \mathbf{F}_{\mathrm{I}}, \mathbf{F}_{\mathrm{v}} \in \mathbb{R}^{3\times 3}$ are positive-definite diagonal gain matrices,
- $\mathbf{v}_{\mathrm{ref}}$ and $\boldsymbol{a}_{\mathrm{ref}}$ are the feedforward velocity and acceleration, often derived from a trajectory generator.

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Position and Velocity Control

Mapping to Thrust

The desired acceleration $\boldsymbol{a}_{\mathrm{d}}$ is then mapped to a desired thrust direction and magnitude. Assuming the only controllable force is along the body z-axis, the total thrust f should satisfy:

$$a_z = g - \frac{f}{m} \quad \Rightarrow \quad f_{\mathrm{d}} = -m(a_{z,\mathrm{d}} - g)$$

where:

- m is the mass of the quadrotor.
- g is the gravity vector.
- f_{d} is the desired total thrust.

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Position and Velocity Control

Attitude Control Command Generation

To achieve the desired thrust direction, the quadrotor's attitude must be controlled so that the body z-axis aligns with the desired acceleration. A nominal yaw angle ψ_d is provided (e.g., from a planner or user input). The full desired roll and pitch angle is then constructed using:

$$\begin{bmatrix} a_x \\ a_y \end{bmatrix} = -g\mathbf{A}_\psi \begin{bmatrix} \phi \\ \theta \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} \phi_\mathrm{d} \\ \theta_\mathrm{d} \end{bmatrix} = -g^{-1}\mathbf{A}_\psi^{-1} \begin{bmatrix} a_{x,\mathrm{d}} \\ a_{y,\mathrm{d}} \end{bmatrix}$$

where:

- $\begin{bmatrix} a_{x,\mathrm{d}} \\ a_{y,\mathrm{d}} \end{bmatrix}$ is the desired x-axis and y-axis acceleration generated by the position controller.
- $\begin{bmatrix} \phi_\mathrm{d} \\ \theta_\mathrm{d} \end{bmatrix}$ is the desired total roll and pitch angle.
- $\mathbf{A}_\psi = \begin{bmatrix} \sin\psi & \cos\psi \\ -\cos\psi & \sin\psi \end{bmatrix}$, which is derived in the model simplification

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Control Allocation

Objective:

The goal of control allocation is to determine the individual rotor inputs $[\delta_1, \delta_2, \delta_3, \delta_4]^T$ that produce a desired total thrust f_d and body torques $\boldsymbol{\tau}_\mathrm{d} = \left[\tau_{\mathrm{x,d}}, \tau_{\mathrm{y,d}}, \tau_{\mathrm{z,d}}\right]^T$, as commanded by the position and attitude controllers.

Control Inputs

The quadrotor is controlled via four inputs:

- Total thrust: f_d
- Body torques: $\tau_\mathrm{d} = \left[\tau_{\mathrm{x,d}}, \tau_{\mathrm{y,d}}, \tau_{\mathrm{z,d}}\right]^T$

These control inputs are generated by the position and attitude control loops and need to be translated into individual motor commands.

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Control Allocation

Allocation Matrix with Steady Motor Gain

Each rotor produces a thrust force and a reactive torque (yaw moment) due to drag. The mapping from rotor inputs to control inputs of the UAV body is given by:

$$\Delta \begin{bmatrix} f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} \Delta f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \cdot 2 \begin{bmatrix} (\delta_1 - \delta_0)/\delta_0 \\ (\delta_2 - \delta_0)/\delta_0 \\ (\delta_3 - \delta_0)/\delta_0 \\ (\delta_4 - \delta_0)/\delta_0 \end{bmatrix} = \frac{\mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0^2 \cdot 2}{\delta_0} \left(\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} - \begin{bmatrix} \delta_0 \\ \delta_0 \\ \delta_0 \\ \delta_0 \end{bmatrix} \right) = \mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0 \cdot K_m \cdot 2 \left(\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} - \begin{bmatrix} \delta_0 \\ \delta_0 \\ \delta_0 \\ \delta_0 \end{bmatrix} \right)$$

Solving for Rotor Inputs

To compute rotor inputs from desired control inputs:

$$\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}_d = (\mathbf{M} \cdot \mathbf{\Gamma} \cdot \Omega_0 \cdot K_m \cdot 2)^{-1} \begin{bmatrix} \Delta f \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix}_d + \delta_0$$

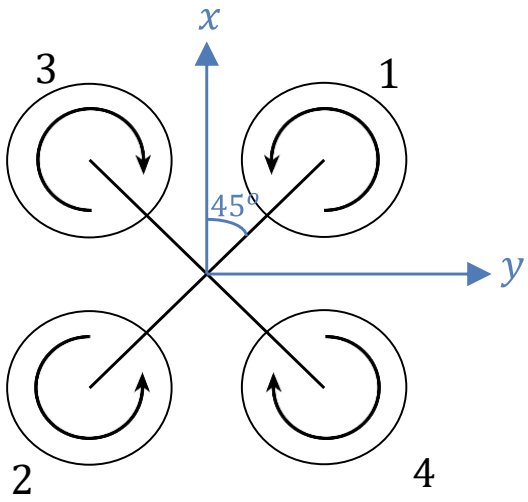
Remember what is M and Γ?

Allocation Matrix

This direct inversion of the matrix is valid under ideal conditions

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Control Allocation



Remarks

- Control allocation ensures physical feasibility and safety by respecting rotor limits.
- Allocation can also be extended to fault-tolerant or over-actuated systems (e.g., hexarotors).
- This direct inversion is valid under ideal conditions. In practice, additional constraints (e.g., actuator saturation, non-negativity of thrusts, delay) are considered, and constrained optimization or pseudo-inverse solutions are used.

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Control Allocation

Simplification in Common Open-source Flight Controller

Normally a **mixing matrix (N)** will be multiplied with the motor inputs to decouple the effects of the motor inputs.

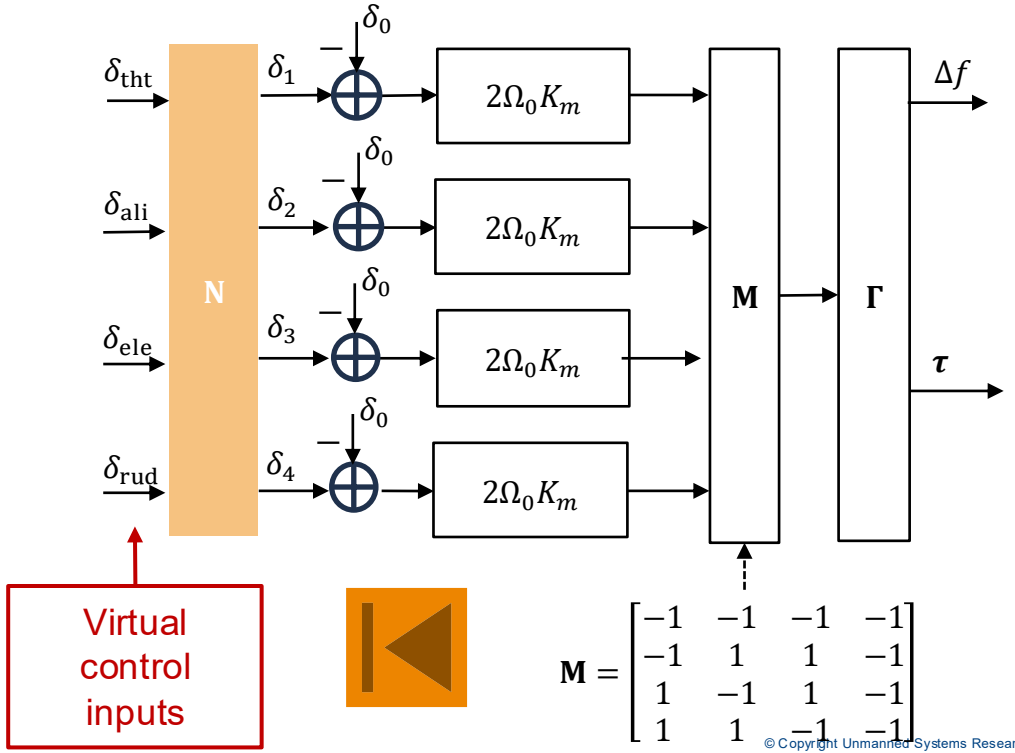
$$\begin{bmatrix}\delta_1\\ \delta_2\\ \delta_3\\ \delta_4\end{bmatrix} = \mathbf{N}\begin{bmatrix}\delta_{\text{tht}}\\ \delta_{\text{ali}}\\ \delta_{\text{ele}}\\ \delta_{\text{rud}}\end{bmatrix}$$

where

$$\mathbf{N} = \begin{bmatrix}-1 & -1 & 1 & 1\\ -1 & 1 & -1 & 1\\ -1 & 1 & 1 & -1\\ -1 & -1 & -1 & -1\end{bmatrix}$$

$\mathbf{N} = 4 \cdot \mathbf{M}^{-1}$

$$\mathbf{M}^{-1} = \begin{bmatrix}-0.25 & -0.25 & 0.25 & 0.25\\ -0.25 & 0.25 & -0.25 & 0.25\\ -0.25 & 0.25 & 0.25 & -0.25\\ -0.25 & -0.25 & -0.25 & -0.25\end{bmatrix}$$



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Control Allocation

Revised Forces and Moments Generation

The revised mapping from rotor inputs to control inputs of the UAV body is given by:

$$\begin{bmatrix}\Delta f\\ \tau_{\mathrm{x}}\\ \tau_{\mathrm{y}}\\ \tau_{\mathrm{z}}\end{bmatrix} = \mathbf{M}\cdot\boldsymbol{\Gamma}\cdot\Omega_0\cdot K_m\cdot 2\left(\begin{bmatrix}\delta_1\\ \delta_2\\ \delta_3\\ \delta_4\end{bmatrix} - \begin{bmatrix}\delta_0\\ \delta_0\\ \delta_0\\ \delta_0\end{bmatrix}\right) = \mathbf{M}\cdot\boldsymbol{\Gamma}\cdot\Omega_0\cdot K_m\cdot 2\left(\mathbf{N}\begin{bmatrix}\delta_{\mathrm{tht}}\\ \delta_{\mathrm{ali}}\\ \delta_{\mathrm{ele}}\\ \delta_{\mathrm{rud}}\end{bmatrix} - \begin{bmatrix}\delta_0\\ \delta_0\\ \delta_0\\ \delta_0\end{bmatrix}\right)$$

$$= 8\cdot\boldsymbol{\Gamma}\cdot\Omega_0\cdot K_m\left(\begin{bmatrix}\delta_{\mathrm{tht}}\\ \delta_{\mathrm{ali}}\\ \delta_{\mathrm{ele}}\\ \delta_{\mathrm{rud}}\end{bmatrix} - \mathbf{N}^{-1}\begin{bmatrix}\delta_0\\ \delta_0\\ \delta_0\\ \delta_0\end{bmatrix}\right) = 8\cdot\boldsymbol{\Gamma}\cdot\Omega_0\cdot K_m\left(\begin{bmatrix}\delta_{\mathrm{tht}}\\ \delta_{\mathrm{ali}}\\ \delta_{\mathrm{ele}}\\ \delta_{\mathrm{rud}}\end{bmatrix} - \begin{bmatrix}\delta_{\mathrm{tht},0}\\ 0\\ 0\\ 0\end{bmatrix}\right)$$

$$\boldsymbol{\Gamma} = \begin{bmatrix} k_T & 0 & 0 & 0\\ 0 & \frac{\sqrt{2}}{2}lk_T & 0 & 0\\ 0 & 0 & \frac{\sqrt{2}}{2}lk_T & 0\\ 0 & 0 & 0 & k_Q\end{bmatrix}$$

Since $\boldsymbol{\Gamma}$ is a diagonal matrix, simplified force and moment generation is decoupled

Solving for the Virtual Control Inputs

To compute rotor inputs from desired control inputs:

$$\begin{bmatrix}\delta_{\mathrm{tht}}\\ \delta_{\mathrm{ali}}\\ \delta_{\mathrm{ele}}\\ \delta_{\mathrm{rud}}\end{bmatrix}_{\mathrm{d}} = (8\cdot\boldsymbol{\Gamma}\cdot\Omega_0\cdot K_m)^{-1}\begin{bmatrix}\Delta f\\ \tau_{\mathrm{x}}\\ \tau_{\mathrm{y}}\\ \tau_{\mathrm{z}}\end{bmatrix}_{\mathrm{d}} + \begin{bmatrix}\delta_{\mathrm{tht},0}\\ 0\\ 0\\ 0\end{bmatrix}$$

In most autopilots, controllers are all PID controllers. So, the unknown variables can be compensated for by adjusting PID parameters

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Control Allocation

Compute the hovering states to compute more accurate throttle input

The virtual control values at the ideal hovering condition can be computed as follows:

$$\begin{bmatrix} \delta_{\text{tht},0} \\ 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{N}^{-1} \begin{bmatrix} \delta_0 \\ \delta_0 \\ \delta_0 \\ \delta_0 \end{bmatrix} = \begin{bmatrix} 0.25 & -0.25 & 0.25 & 0.25 \\ -0.25 & 0.25 & 0.25 & -0.25 \\ 0.25 & -0.25 & 0.25 & -0.25 \\ 0.25 & 0.25 & -0.25 & -0.25 \end{bmatrix} \begin{bmatrix} \delta_0 \\ \delta_0 \\ \delta_0 \\ \delta_0 \end{bmatrix}$$

Thus,

$$\delta_{\text{tht},0} = \delta_0$$

where,

$$\delta_0 = \frac{\Omega_0}{K_m} = \frac{\Omega_0^2}{K_m \Omega_0} = \frac{\frac{mg}{4k_T}}{K_m \Omega_0} = \frac{mg}{4k_T \Omega_0 K_m}$$

The throttle input at hover is an important parameter in flight control, and it can be also accurately determined through flight tests.

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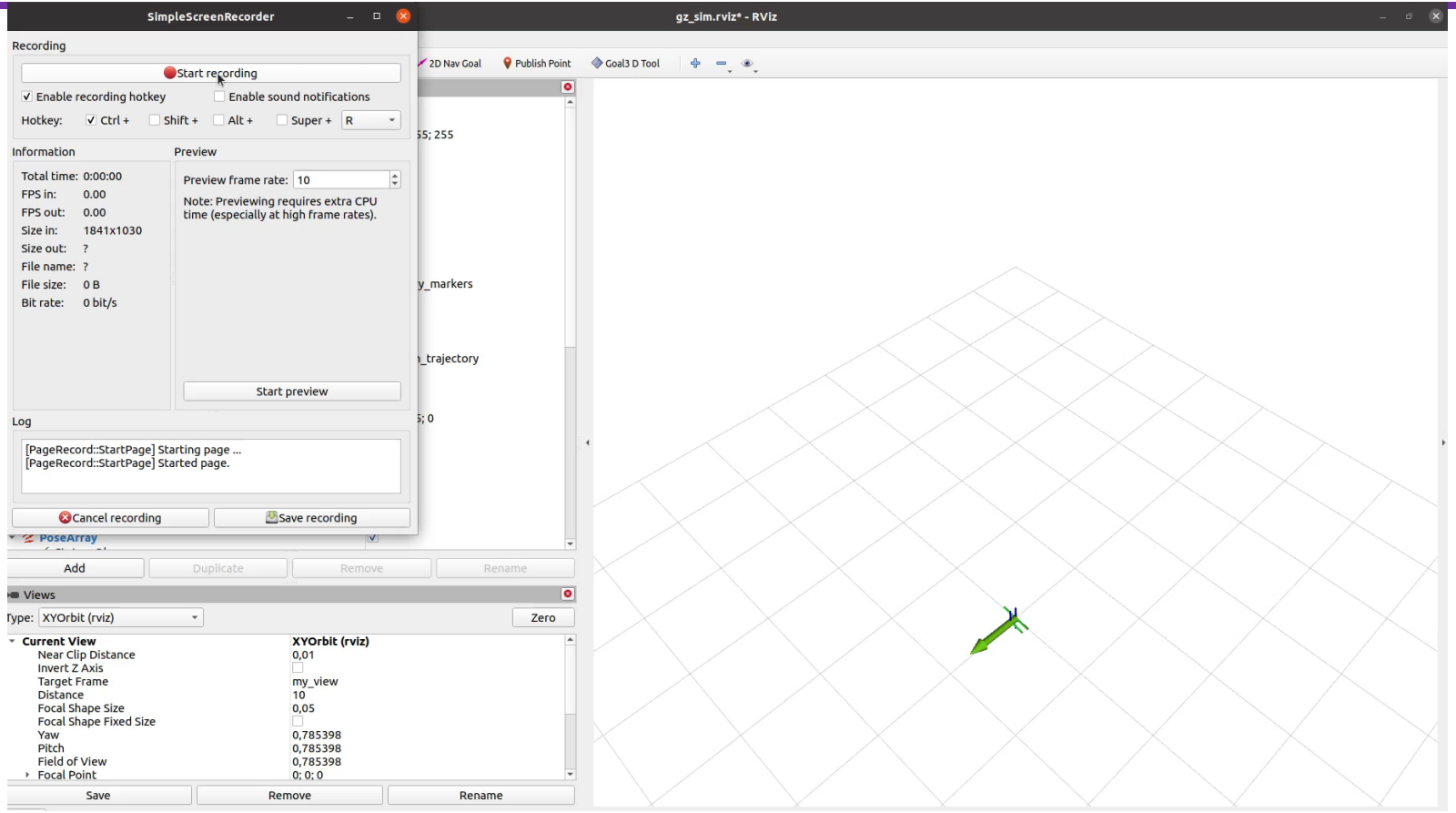


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Simulation



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Summary

- The controller design for multicopters is based on specific control requirements. For different control requirements, different mathematical models and corresponding control methods are selected.
- The design using successive loop closure is often adopted in flight control. Multiple closed-loop feedbacks work together to realize the flight control of multicopters.
- Low-level flight control of multicopters mostly depends on the PID control, which is sufficient for most requirements. However, it is not enough for tough requirements. Therefore, a question is being asked: how to design low-level flight controllers for multicopters to obtain a stability region as large as possible, decrease vibration, save energy, resist wind, achieve a high precision hover or an aggressive maneuver flight?

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Thank You!

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